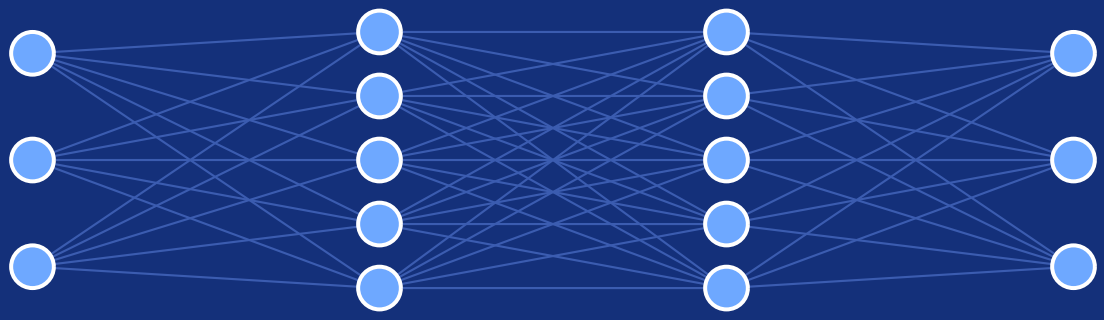


AI OLYMPIAD

CLASS 10 STUDENT BOOK

Content | Activities | Projects | Worksheets | Answer Key



Senior Level (Class 6-10) | 6-Month Course | October to April

6 chapters x 25 pages

Student Name : _____

Class / Section : _____

School : _____

Mentor : _____

ABOUT THIS BOOK

How to use it, week by week

Welcome, AI Explorer!

This book has 6 chapters, one for each month of your course. Every chapter has exactly 25 pages and follows the same simple flow, so you always know what to do next.

LEARN
Read 6 lessons with diagrams.

DO
Try 5 team activities.

PRACTISE
Mini project and 2 worksheets.

REVISE
Puzzle, glossary and self-check.

What is inside each 25-page chapter

Page 1	Chapter opener: illustration, goals, key words
Page 2	Warm-up: KWL chart, guess and write, chapter roadmap
Pages 3-14	6 lessons (2 pages each): Learn page with diagram and mind map, Practise page with 5 questions
Pages 15-19	5 activity pages with result tables
Pages 20-21	Mini project: plan, build, test, present
Pages 22-23	Worksheet A and Worksheet B
Page 24	Revision in 10 points and word search
Page 25	Glossary, I-can checklist and reflection

Golden rules for using AI

- Use AI to learn, not to copy.
- Check AI answers with your textbook or trusted sources.
- Never share private details like phone number, address or ID numbers.
- Tell your teacher when AI helped you.
- Ask when you are not sure.

Monthly assessment (20 marks)

Criteria	Marks	What to look for
Concept understanding	5	Explains ideas in own words with examples
Activity participation	5	Teamwork, effort, good questions
Worksheet and project	5	Correct answers, neat working, tested prototype
Creativity and responsibility	5	Original ideas, honest and safe use of AI

Note: tools such as Teachable Machine and online Python editors are suggestions. Check availability and your school's AI-use policy. Activities can also be done offline with printed cards.

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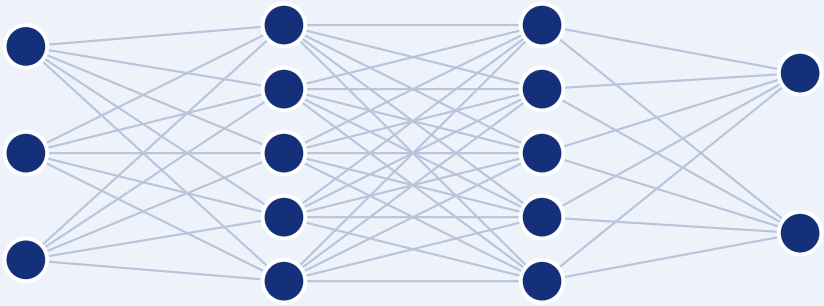
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CHAPTER 1 | MONTH 1 | AI BASICS

How AI Thinks with Data | Generative AI Uncovered

DATA > PATTERNS > PREDICTIONS



Input data | Learning layers | Prediction

In this chapter you will

- Explain what AI is and how it differs from normal software.
- Describe how data helps machines learn patterns and make predictions.
- Explain how generative AI and Large Language Models create content, and know their limits.

Key words

Data	Facts, numbers, text, images or sounds that AI learns from.
Pattern	A repeated feature or rule found in data.
Model	The trained 'brain' of an AI system that makes predictions.
Machine Learning	Learning from examples instead of fixed rules.
Generative AI	AI that creates new text, images, audio or code.
LLM	Large Language Model, trained on huge amounts of text.
Hallucination	A confident but false answer given by AI.

Did you know?
The term 'artificial intelligence' was first used for a research workshop at Dartmouth College, USA, in 1956.

WARM-UP | THINK BEFORE YOU LEARN

Chapter 1: AI Basics

Before you start this chapter, find out what you already know. There are no wrong answers here. Guess, think and write.

My KWL chart

K - What I already KNOW	W - What I WANT to know	L - What I LEARNED (fill at the end)

Guess and write

1. Where have you seen a computer or phone 'guess' something correctly, like your next word or a video you may like?

_____

_____

2. Can a machine think like a human? Write your opinion.

_____

_____

3. What do you think a computer needs before it can recognise a cat in a photo?

_____

_____

4. If AI makes a mistake, who should check it?

_____

_____

5. Write one thing you hope to learn in this chapter.

_____

_____

Chapter roadmap

Lesson 1.1
What is Artificial Intelligence?

Lesson 1.2
Data: The Food of AI

Lesson 1.3
How Machines Learn

Lesson 1.4
How AI Makes Choices

Lesson 1.5
Generative AI Uncovered

Lesson 1.6
LLMs, Limits and Hallucinations

LESSON 1.1 | WHAT IS ARTIFICIAL INTELLIGENCE?

Chapter 1: AI Basics - Learn

Artificial Intelligence (AI) means machines or software that can do tasks which normally need human intelligence. AI has no feelings or body like us. It works with data and instructions.

Intelligence	Humans learn, understand language, solve problems and make decisions. AI tries to copy some of these abilities.
Narrow AI	Most AI today is narrow. It is good at one job, such as recognising faces or translating text, but cannot do everything a person can.
AI in daily life	Voice assistants, map apps, video recommendations, spam filters and phone face unlock all use AI.
AI vs normal software	Normal software follows fixed rules written by a programmer. AI learns patterns from examples and can handle new situations.

Normal software	AI system
• Follows fixed rules • Programmer writes every step • Same input gives same output • Example: calculator	• Learns from examples • Finds patterns in data • Can handle new cases • Example: face unlock

Real-life example
A calculator always adds 2 + 2 with a fixed rule. A spam filter was never told every spam email. It learned from thousands of examples what spam usually looks like.

Lesson at a glance



Discuss with a partner: where do you see 'What is Artificial Intelligence?' in real life? Write your best example.

_____

_____

LESSON 1.1 | PRACTISE: WHAT IS ARTIFICIAL INTELLIGENCE?

Chapter 1: AI Basics

Try it

Task
List 5 places where you met AI today (home, phone, school, road, shop). For each one, write what the AI did.

Check your understanding

1. Define Artificial Intelligence in your own words.

2. What is narrow AI?

3. Give two differences between normal software and AI.

4. Name three AI tools or apps you use at home.

5. Can AI feel happy or sad? Explain.

Remember
- AI copies some human abilities using data.
- Most AI today is narrow AI.
- AI learns from examples; normal software follows fixed rules.

LESSON 1.2 | DATA: THE FOOD OF AI

Chapter 1: AI Basics - Learn

Data is the raw material of AI, like food for our body. Without data an AI system has nothing to learn from. Good data helps build good AI.

Types of data	Text (messages), images (photos), audio (voice), video and numbers (marks, temperature) are all data.
Structured and unstructured	Structured data fits into tables, like a marks list. Unstructured data, such as photos and voice notes, has no fixed table shape.
Good data	Good data is correct, large enough, varied and up to date.
Bad data and bias	If data is wrong, missing or one-sided, AI gives poor or unfair results. Remember: garbage in, garbage out.

Data type	Example	AI task
Text	Chat messages	Spam detection, translation
Image	Photos of leaves	Plant disease detection
Audio	Voice command	Speech recognition
Numbers	Daily temperature	Weather prediction

Real-life example
A face-unlock system trained on photos of only a few kinds of faces may fail on others. The cause is one-sided data, not bad intention.

Lesson at a glance



Discuss with a partner: where do you see 'Data: The Food of AI' in real life? Write your best example.

_____

_____

LESSON 1.2 | PRACTISE: DATA: THE FOOD OF AI

Chapter 1: AI Basics

Try it

Task
Collect data: ask 15 classmates how they come to school (walk, cycle, bus, bike, car). Make a small table and write one pattern you notice.

Check your understanding

1. What is data?

2. Name four types of data.

3. Write one difference between structured and unstructured data with an example of each.

4. What does 'garbage in, garbage out' mean?

5. Why is varied data important for AI?

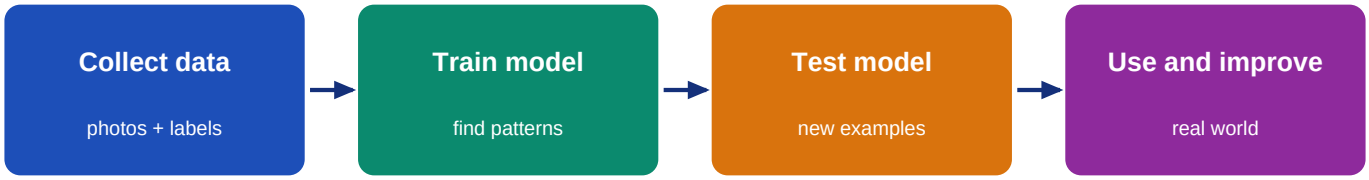
Remember
- Data is what AI learns from.
- Data can be text, images, audio, video or numbers.
- Poor or one-sided data gives poor or unfair AI.

LESSON 1.3 | HOW MACHINES LEARN

Chapter 1: AI Basics - Learn

Machine Learning (ML) is a part of AI where a machine learns from examples instead of being told every rule. It improves with experience, just as you improve with practice.

Training	The model is shown many examples, such as photos of cats with the label 'cat'. It adjusts itself to find patterns.
Testing	After training, the model is tested on new examples it has not seen, to check how accurate it is.
Supervised learning	Learns from labelled examples. Example: emails labelled 'spam' or 'not spam'.
Unsupervised and reinforcement	Unsupervised learning groups unlabelled data, like grouping shoppers by habits. Reinforcement learning learns by trial, with rewards and penalties, like a game-playing AI.



Real-life example
Before an exam your teacher gives 20 solved examples. You find the pattern and solve a new question. A machine learning model does the same, but with thousands of examples.

Lesson at a glance



Discuss with a partner: where do you see 'How Machines Learn' in real life? Write your best example.

_____

_____

LESSON 1.3 | PRACTISE: HOW MACHINES LEARN

Chapter 1: AI Basics

Try it

Task
Play 'Teach the Robot'. Write 3 rules that help a robot tell a pen from a pencil. Then write one object for which your rules may fail.

Check your understanding

1. What is machine learning?

2. Why do we test a model with new examples?

3. What is supervised learning? Give an example.

4. How is reinforcement learning like training a pet?

5. Which type needs labels: supervised or unsupervised learning?

Remember
- ML learns from examples.
- Models are trained and then tested on new data.
- Three types: supervised, unsupervised and reinforcement.

LESSON 1.4 | HOW AI MAKES CHOICES

Chapter 1: AI Basics - Learn

After learning, an AI model uses what it found to make choices. These choices are usually predictions based on probability, not magic and not certainty.

Classification	Putting something into a group. Example: is this email spam or not spam?
Prediction	Estimating a number or a future event. Example: tomorrow's temperature or a bus arrival time.
Recommendation	Suggesting things you may like based on patterns, as video and music apps do.
Probability and accuracy	AI often gives a confidence level, like 86 percent sure. No model is 100 percent correct, so humans must check important results.

Confidence of an image model for one photo



Real-life example
A phone camera app says a photo shows 'food' with 95 percent confidence. If the photo is blurry, the confidence drops. The model is estimating, not knowing.

Lesson at a glance



Discuss with a partner: where do you see 'How AI Makes Choices' in real life? Write your best example.

_____

_____

LESSON 1.4 | PRACTISE: HOW AI MAKES CHOICES

Chapter 1: AI Basics

Try it

Task
Write one example each of classification, prediction and recommendation that you have seen in an app you use.

Check your understanding

1. What is classification? Give an example.

2. What is the difference between classification and prediction?

3. How do music or video apps recommend things to you?

4. What does 'confidence level' of an AI model mean?

5. Why should humans check important AI results?

Remember
- AI makes classifications, predictions and recommendations.
- AI results are based on probability.
- No AI model is perfect, so humans must check.

LESSON 1.5 | GENERATIVE AI UNCOVERED

Chapter 1: AI Basics - Learn

Generative AI creates new content such as text, images, music, video or computer code. It learns patterns from huge amounts of existing examples and then produces something new that follows those patterns.

Text generation	Chat assistants answer questions, summarise notes and write drafts.
Image and audio generation	Tools can create pictures or voices from a written description called a prompt.
Code generation	Coding assistants suggest and explain code, but a human must test it.
Traditional AI vs generative AI	Traditional AI mainly analyses or classifies existing data. Generative AI creates new content.

Traditional AI	Generative AI
• Analyses data • Classifies and predicts • Output: label or number • Example: spam filter	• Creates new content • Writes, draws, composes • Output: text, image, audio, code • Example: chat assistant

Real-life example
If you give an image tool the prompt 'a blue elephant flying over a school', it combines patterns it learned about elephants, colours and schools to make a brand new picture.

Lesson at a glance



Discuss with a partner: where do you see 'Generative AI Uncovered' in real life? Write your best example.

_____

_____

LESSON 1.5 | PRACTISE: GENERATIVE AI UNCOVERED

Chapter 1: AI Basics

Try it

Task
Write a prompt for an AI tool to create a poster idea for 'Save Water'. Then list 3 things you would check in the result.

Check your understanding

1. What is generative AI?

2. Name three kinds of content generative AI can create.

3. What is a prompt?

4. How is generative AI different from a spam filter?

5. Why must code written by AI be tested?

Remember
- Generative AI creates new content from patterns.
- A prompt is your instruction to the AI.
- Always check and test AI output.

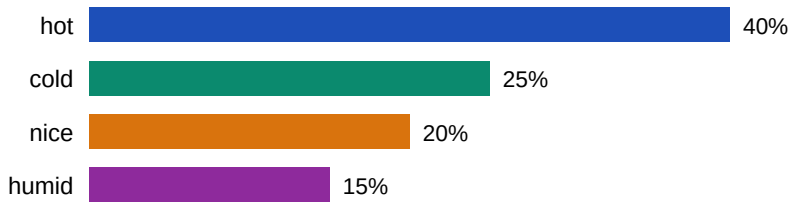
LESSON 1.6 | LLMS, LIMITS AND HALLUCINATIONS

Chapter 1: AI Basics - Learn

A Large Language Model (LLM) is a generative AI trained on a very large amount of text. It is the technology behind chat assistants. It is powerful, but it has limits that every user must know.

Tokens	LLMs split text into small pieces called tokens. A token can be a word or part of a word.
Next-token prediction	An LLM predicts the most likely next token again and again to build sentences and paragraphs.
Hallucination	Sometimes it writes confident but false statements, such as a book or a fact that does not exist.
Other limits	It can show bias from its training data, may not know recent events, and does not understand the world like a human. Always verify.

Prompt: "The weather today is very ..." (illustrative numbers)



LLM picks a likely next word, then repeats.

Real-life example
A student asks an AI tool for the title of a book on a topic. The tool gives a title that sounds real but does not exist. Checking in a library catalogue reveals the hallucination.

Lesson at a glance



Discuss with a partner: where do you see 'LLMs, Limits and Hallucinations' in real life? Write your best example.

_____

_____

LESSON 1.6 | PRACTISE: LLMS, LIMITS AND HALLUCINATIONS

Chapter 1: AI Basics

Try it

Task
Ask a classmate or an AI tool one factual question you already know the answer to. Write the answer and mark if it is correct. What did you learn?

Check your understanding

1. What does LLM stand for?

2. What is a token?

3. How does an LLM build a sentence?

4. What is a hallucination? Give an example.

5. List two ways to reduce the risk of believing wrong AI answers.

Remember
- LLMs predict the next token using patterns.
- They can hallucinate and show bias.
- Always verify important information.

ACTIVITY 1.1 | HUMAN CLASSIFIER

Chapter 1: AI Basics - Do

Time: 20 min **Group:** Pairs **Materials:** 20 picture cards (fruits, vehicles, animals), pencil

Goal

Act as a machine learning model: learn rules from labelled examples and test them on new examples.

Steps

1. Student A is the 'AI'. Student B shows 10 labelled cards (training data) one by one.

2. Student A writes the rules they notice (colour, shape, size, parts).

3. Student B now shows 5 new cards without labels (test data).

4. Student A predicts the label for each card using the rules.

5. Count how many predictions are correct and calculate accuracy.

Record your results

Card shown	AI's prediction	Correct? (Yes / No)

Think and write

Which card confused the 'AI'? What extra training data would have helped?

_____

_____

_____

_____

ACTIVITY 1.2 | PREDICT THE NEXT WORD

Chapter 1: AI Basics - Do

Time: 15 min **Group:** Whole class **Materials:** Whiteboard, notebook

Goal

Understand how a language model predicts the next word using probabilities.

Steps

1. The teacher writes: 'Today the weather is very ...'.

2. Each student writes the first word that comes to mind.

3. Count how many students wrote each word.

4. Write the counts as percentages.

5. Repeat with: 'My favourite subject is ...' and 'After school I go to ...'.

Record your results

Sentence	Most common word	Percent of class

Think and write

An LLM does this with billions of sentences. Why do you think it sometimes gets a sentence wrong?

_____

_____

_____

_____

ACTIVITY 1.3 | DATA DETECTIVE

Chapter 1: AI Basics - Do

Time: 30 min **Group:** Groups of 4 **Materials:** Survey sheet, pencils, chart paper

Goal

Collect data, find patterns and notice bias in data.

Steps

1. Choose a question, such as 'Which snack do students like best?'.

2. Ask 10 students from your own class and record the answers.

3. Now ask 10 students from a different class and record again.

4. Make a bar chart for each set.

5. Compare the two charts and look for differences.

Record your results

Option	Count: my class	Count: other class

Think and write

If an AI learned only from your class data, what could go wrong? How does this show bias?

_____

_____

_____

_____

ACTIVITY 1.4 | TRAIN A TINY IMAGE MODEL

Chapter 1: AI Basics - Do

Time: 35 min **Group:** Groups of 3 **Materials:** Computer with webcam, internet, Google Teachable Machine

Goal

Train a simple image classifier and see how data quality changes accuracy.

Steps

1. Open Teachable Machine and choose an Image Project.

2. Create two classes, for example 'Pen' and 'Eraser'.

3. Capture about 30 images of each class from different angles.

4. Train the model and test it with new objects.

5. Retrain with only 5 images or a messy background and compare results.

Record your results

Trial	Number of images	What happened

Think and write

What happened when you used fewer or poorer images? Link your answer to 'garbage in, garbage out'.

_____

_____

_____

_____

ACTIVITY 1.5 | HALLUCINATION HUNT

Chapter 1: AI Basics - Do

Time: 25 min **Group:** Pairs **Materials:** Textbook, AI-generated answers prepared by the teacher

Goal

Learn to verify AI answers and catch hallucinations.

Steps

1. The teacher gives 5 short AI-written answers with a few hidden mistakes.

2. Read each answer carefully.

3. Check each fact with your textbook or a trusted source.

4. Mark each answer as correct, partly correct or wrong.

5. Rewrite one wrong answer correctly.

Record your results

Answer no.	Correct / Wrong	Source I used to check

Think and write

Which kind of mistake was hardest to spot? What will you do before trusting an AI answer?

_____

_____

_____

_____

MINI PROJECT | PART 1: PLAN

Chapter 1: AI Basics

My First Smart Classifier

Build a small classifier that can tell two things apart, then explain how it learned, how accurate it is and where it fails. You can do it with paper cards (human model) or with a free no-code tool.

Pick an idea (tick one)

[ ] Pen versus pencil classifier
[ ] Healthy leaf versus damaged leaf classifier
[ ] Hand sign classifier (open palm versus fist)
[ ] Recyclable versus non-recyclable waste classifier

Project plan

Plan	My answer
Project name	
Problem I am solving	
Who will use it	
Tools / AI I will use	
Data or materials needed	
Team members and roles	
Time I need	

Success criteria

- Uses at least two classes with 20 or more examples each (or 10 cards).
- Records training data and test results in a table.
- Shows accuracy and one case where the model failed.
- Explains how better data would improve the result.

MINI PROJECT | PART 2: BUILD, TEST, PRESENT

Chapter 1: AI Basics

Build tip

Collect varied examples (different light, angles, backgrounds). Keep some examples aside only for testing. Do not use photos of people without their permission.

Test log

Test	What I tried	Result	What I will improve
1			
2			
3			
4			

Presentation checklist (tick when done)

[ ] I can explain the problem in one sentence.
[ ] I can show how my project works.
[ ] I can say where AI or data is used.
[ ] I can name one limit or risk.
[ ] I checked facts and privacy.
[ ] I thanked my team and mentor.

Peer feedback: 2 stars and a wish

Star 1	Star 2	Wish (one improvement)

Project score (teacher)

Idea and purpose	Working prototype	Responsible AI use	Presentation	Total
5	5	5	5	/ 20

WORKSHEET A | CHAPTER 1: AI BASICS

Name: _____ Class: _____ Date: _____ Marks: ____ / 18

Section A - Choose the correct answer (8 x 1 = 8)

1. In machine learning, a model mainly learns from:
   (a) fixed rules typed by a programmer
   (b) examples in data
   (c) the speed of the internet
   (d) the size of the screen

2. 'Garbage in, garbage out' means:
   (a) AI cleans bad data by itself
   (b) poor data leads to poor results
   (c) garbage can train AI
   (d) output is always better than input

3. Which of these is generative AI?
   (a) A calculator
   (b) A tool that writes a story from a prompt
   (c) An alarm clock
   (d) A torch

4. An LLM produces text mainly by:
   (a) copying a stored answer
   (b) predicting the next token using learned patterns
   (c) asking a human typist
   (d) searching only a dictionary

5. Most AI used today is:
   (a) general AI that can do everything
   (b) narrow AI that does specific tasks
   (c) a robot with feelings
   (d) human intelligence

6. Labelled examples are used in:
   (a) supervised learning
   (b) no type of learning
   (c) only games
   (d) only graphics

7. A photo is an example of:
   (a) structured data
   (b) unstructured data
   (c) a rule
   (d) a program

8. The first use of the term 'artificial intelligence' for a workshop was in:
   (a) 1856
   (b) 1956
   (c) 2006
   (d) 2020

Section B - Fill in the blanks (6 x 1 = 6)

1. AI finds _____ in data to make predictions.

2. When AI gives a confident but false answer, it is called a _____.

3. Data that already has answers (labels) is used in _____ learning.

4. A small piece of text processed by an LLM is called a _____.

5. After training, a model is _____ with new examples.

6. Unfair results caused by one-sided data are called _____.

Section C - True or False (4 x 1 = 4)

1. Normal software learns from examples without any rules. **(True / False)**

2. A model should be tested on new examples. **(True / False)**

3. AI is always 100 percent correct. **(True / False)**

4. Generative AI can create new images. **(True / False)**

WORKSHEET B | CHAPTER 1: AI BASICS

Name: _____ Class: _____ Date: _____ Marks: ____ / 24

Section D - Match the columns (5 x 1 = 5)

Column A	Answer	Column B
1. Data	_____	(a) Instruction given to an AI tool
2. Model	_____	(b) Learning by rewards and penalties
3. Prompt	_____	(c) Facts, text, images or numbers AI learns from
4. Hallucination	_____	(d) Confident but false AI answer
5. Reinforcement learning	_____	(e) The trained 'brain' that makes predictions

Section E - Short answers (4 x 3 = 12)

1. Explain with an example how an email spam filter learns to identify spam.

2. Write two limitations of generative AI.

3. Differentiate between classification and prediction with one example each.

4. Why is data quality important for an AI model?

Section F - Practical task (7)

Choose 3 features you would give an AI to tell a cat from a dog. Then write one case where these features might fail. Explain how more data could help.

REVISION AND PUZZLE

Chapter 1: AI Basics

This chapter in 10 points

1. AI means machines doing tasks that normally need human intelligence.

2. Most AI today is narrow AI built for one job.

3. Data (text, images, audio, video, numbers) is what AI learns from.

4. Garbage in, garbage out: poor data gives poor results.

5. Machine learning learns from examples instead of fixed rules.

6. Models are trained and then tested on new data.

7. Three learning types: supervised, unsupervised, reinforcement.

8. AI makes classifications, predictions and recommendations using probability.

9. Generative AI creates new text, images, audio and code from patterns.

10. LLMs predict the next token, can hallucinate and show bias, so always verify.

Word search: find these 8 words (across, down or diagonal)

M	L	M	P	C	F	O	M	S	P	I	E
N	S	E	I	T	O	K	E	N	A	Y	N
L	W	M	A	H	E	T	C	F	T	E	H
W	H	A	P	R	T	R	F	I	T	J	A
E	N	S	L	U	N	A	T	K	E	E	Y
D	R	U	V	W	B	I	O	W	R	S	M
M	A	M	M	D	P	N	N	V	N	M	B
G	C	T	G	O	F	I	D	G	M	K	U
B	D	A	A	T	E	N	S	D	O	L	U
P	R	O	M	P	T	G	A	C	D	G	U
M	E	V	I	L	U	L	P	D	E	D	P
O	P	P	B	I	A	S	J	C	L	E	D

DATA | MODEL | TOKEN | PATTERN | BIAS | TRAINING | PROMPT | LEARNING

GLOSSARY AND SELF-CHECK

Chapter 1: AI Basics

Glossary

Data	Facts, numbers, text, images or sounds that AI learns from.
Pattern	A repeated feature or rule found in data.
Model	The trained 'brain' of an AI system that makes predictions.
Machine Learning	Learning from examples instead of fixed rules.
Generative AI	AI that creates new text, images, audio or code.
LLM	Large Language Model, trained on huge amounts of text.
Hallucination	A confident but false answer given by AI.
Training	Showing a model many examples so it can find patterns.
Token	A small piece of text, such as a word or part of a word.
Bias	Unfair results caused by one-sided or incomplete data.

I can ... (tick the box)

☐	Explain what AI is in simple words.	Yes / Need help
☐	Name different types of data.	Yes / Need help
☐	Describe training and testing of a model.	Yes / Need help
☐	Give examples of classification, prediction and recommendation.	Yes / Need help
☐	Explain what generative AI does.	Yes / Need help
☐	Explain what an LLM is.	Yes / Need help
☐	Spot a possible hallucination.	Yes / Need help
☐	Say why data quality matters.	Yes / Need help

My reflection

The most interesting thing I learned in this chapter was ...

_____

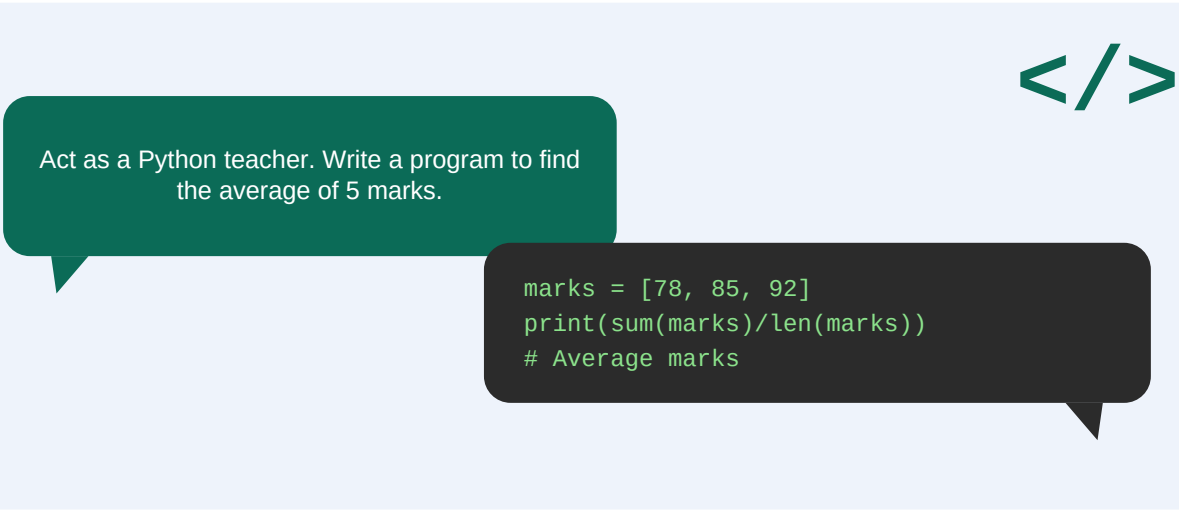
_____

_____

Teacher's remarks: _____ Signature: _____ Date: _____

CHAPTER 2 | MONTH 2 | COMMUNICATE WITH AI

AI-Assisted Coding | Code, Test, Improve



In this chapter you will

- Write clear prompts using Role, Task, Context, Format and Constraints.
- Use basic Python and AI help to write, read and understand small programs.
- Follow the Code, Test, Improve cycle and debug errors responsibly.

Key words

Prompt	The instruction you give to an AI tool.
Role	Who you want the AI to act as.
Context	Background that helps the AI understand your need.
Format	How you want the answer to look, such as a table or list.
Constraint	A limit, such as 'under 50 words'.
Edge case	An unusual input such as 0, negative numbers or empty data.
Bug	A mistake in a program.

Did you know?
Python is named after the British comedy group Monty Python, not the snake.

WARM-UP | THINK BEFORE YOU LEARN

Chapter 2: Communicate with AI

Before you start this chapter, find out what you already know. There are no wrong answers here. Guess, think and write.

My KWL chart

K - What I already KNOW	W - What I WANT to know	L - What I LEARNED (fill at the end)

Guess and write

1. Have you ever asked a voice assistant something and got a wrong answer? What might have gone wrong?

_____

_____

2. How would you explain a new game to a friend who has never played it?

_____

_____

3. What is a program? Write your own definition.

_____

_____

4. Why do you think computers need exact instructions?

_____

_____

5. Write one thing you hope to build with Python or AI.

_____

_____

Chapter roadmap

Lesson 2.1
What is a Prompt?

Lesson 2.2
The 5-Part Prompt

Lesson 2.3
Better Prompting Techniques

Lesson 2.4
Python Basics for AI Coding

Lesson 2.5
Code, Test, Improve

Lesson 2.6
Debugging and Responsible Coding

LESSON 2.1 | WHAT IS A PROMPT?

Chapter 2: Communicate with AI - Learn

A prompt is the instruction, question or request you give to an AI tool. The AI can only work with what you tell it, so the quality of your prompt strongly affects the quality of the answer.

Prompt	Any text you type to an AI, such as a question, a task or an example to follow.
Vague prompt	'Tell me about space' is vague. The AI must guess what you want, so the answer may not help you.
Clear prompt	'Explain black holes to a Class 10 student in 5 simple points' tells the AI the topic, level and format.
Prompt engineering	Writing and improving prompts to get better results is called prompt engineering. It is a skill you can practise.

Vague prompt	Clear prompt
• Write code • Tell me about AI • Help with maths • Result: guessy and long	• Write Python code to add two numbers • Explain AI to a Class 10 student in 5 points • Explain how to solve x + 5 = 12 step by step • Result: focused and useful

Real-life example
Asking a shopkeeper for 'something to write' may get you a crayon. Asking for 'a blue gel pen, 0.5 mm' gets exactly what you need. AI works the same way.

Lesson at a glance



Discuss with a partner: where do you see 'What is a Prompt?' in real life? Write your best example.

_____

_____

LESSON 2.1 | PRACTISE: WHAT IS A PROMPT?

Chapter 2: Communicate with AI

Try it

Task
Take the vague prompt 'Tell me about plants'. Rewrite it in 3 different clear ways for 3 different needs (school project, quiz, simple poster).

Check your understanding

1. What is a prompt?

2. Why can a vague prompt give a poor answer?

3. Write one vague and one clear prompt on the same topic.

4. What is prompt engineering?

5. Can AI read your mind? Explain.

Remember
- A prompt is your instruction to AI.
- Clear prompts give better answers.
- Prompting is a skill that improves with practice.

LESSON 2.2 | THE 5-PART PROMPT

Chapter 2: Communicate with AI - Learn

A strong prompt can be built from five parts: Role, Task, Context, Format and Constraints. You can remember them as RTCFC. You do not always need all five, but they help a lot.

R - Role	Tell the AI who to act as: 'Act as a Python teacher'.
T - Task	State exactly what to do: 'Write a program to find the average of 5 marks'.
C - Context	Give background: 'for a Class 10 beginner who knows only print()'.
F - Format and C - Constraints	Format: 'show the code and then a 3-line explanation'. Constraints: 'use simple words, under 12 lines'.

R	T	C	F	C
Role	**Task**	**Context**	**Format**	**Constraints**
Act as a Python teacher	*Write a program to find the average of 5 marks*	*For a Class 10 beginner*	*Add a comment on each line*	*Use simple words, under 10 lines*

Real-life example
Prompt: 'Act as a friendly science teacher. Explain photosynthesis to a Class 6 student. Use an example from a kitchen garden. Give 5 bullet points under 15 words each.' This uses all five parts.

Lesson at a glance



Discuss with a partner: where do you see 'The 5-Part Prompt' in real life? Write your best example.

_____

_____

LESSON 2.2 | PRACTISE: THE 5-PART PROMPT

Chapter 2: Communicate with AI

Try it

Task
Write a full 5-part prompt asking AI to create a 5-question quiz on a subject of your choice. Label each part R, T, C, F, C.

Check your understanding

1. What does RTCFC stand for?

2. Which part tells the AI who to act as?

3. Give an example of a constraint.

4. Why is context useful?

5. Rewrite 'Make a game' as a 5-part prompt.

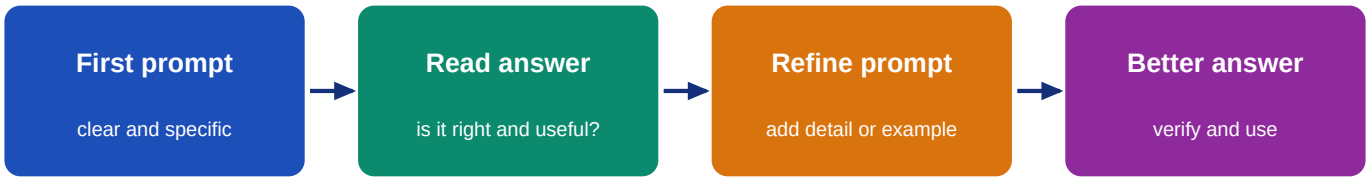
Remember
- RTCFC: Role, Task, Context, Format, Constraints.
- Each part removes guesswork for the AI.
- You can add parts step by step.

LESSON 2.3 | BETTER PROMPTING TECHNIQUES

Chapter 2: Communicate with AI - Learn

Even a good first prompt may not give a perfect answer. Skilled users improve the result with follow-up prompts, examples and step-by-step requests.

Give examples	Show the AI the style you want: 'Write questions like this: What is 2 + 3? (a) 4 (b) 5'.
Ask for steps	Say 'explain step by step' to understand how an answer is reached, not just the answer.
Follow up and refine	If an answer is too long or hard, reply 'make it simpler' or 'use a real-life example'.
Ask for checks	Ask the AI to 'list any assumptions' or 'tell me what you are unsure about'. Then verify yourself.



Real-life example
A student asks: 'Explain fractions.' The answer is too hard. The student follows up: 'Explain with pizza slices for a Class 5 child.' The second answer is far better.

Lesson at a glance



Discuss with a partner: where do you see 'Better Prompting Techniques' in real life? Write your best example.

_____

_____

LESSON 2.3 | PRACTISE: BETTER PROMPTING TECHNIQUES

Chapter 2: Communicate with AI

Try it

Task
Ask an AI tool (or a classmate) to explain 'what is a variable' in two ways: first simply, then with a real-life example. Compare both answers.

Check your understanding

1. How do examples in a prompt help the AI?

2. Why ask the AI to explain step by step?

3. What can you say if an answer is too difficult?

4. Why should you still verify AI answers?

5. What is a follow-up prompt?

Remember
- Examples show the style you want.
- Follow-up prompts refine answers.
- Always verify what AI says.

LESSON 2.4 | PYTHON BASICS FOR AI CODING

Chapter 2: Communicate with AI - Learn

Python is a simple and popular programming language. AI tools can write Python for you, but you must understand it to check and improve it. Here are the basics.

print() and input()	print() shows output on screen. input() takes text typed by the user.
Variables	A variable stores a value, like marks = 85. Use clear names.
if and else	Used to make decisions: if marks >= 35 then pass, else fail.
Loops and lists	A for loop repeats steps. A list stores many values, such as marks = [78, 85, 92].

Python code	What it does
print("Hi")	Shows Hi on screen
x = 5	Stores 5 in variable x
for i in range(3):	Repeats a block 3 times
marks = [70, 80]	Makes a list of two values

```
name = input("Enter your name: ")
marks = int(input("Enter your marks: "))
if marks >= 35:
    print(name, "passed")
else:
    print(name, "needs to improve")
```

Real-life example
A variable is like a labelled tiffin box: the label is the name (marks) and what is inside is the value (85). You can open the box and change what is inside.

Lesson at a glance



Discuss with a partner: where do you see 'Python Basics for AI Coding' in real life? Write your best example.

_____

_____

Try it

Task
Read the program above. What will it print if the name is Anu and the marks are 30? What if the marks are 35? Write both outputs.

Check your understanding

1. What does print() do?

2. What is a variable? Give an example.

3. Why do we use if and else?

4. What is the output of print(3 * 4)?

5. What is a list? Write a list of 3 numbers.

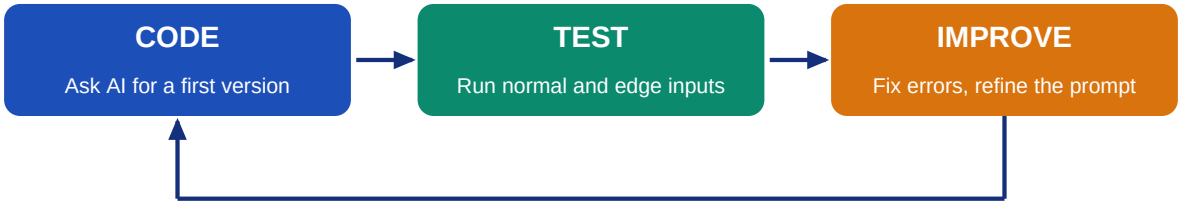
Remember
- print() shows output, input() takes input.
- Variables store values.
- if/else make decisions; loops repeat steps.

LESSON 2.5 | CODE, TEST, IMPROVE

Chapter 2: Communicate with AI - Learn

AI can write a first version of a program quickly. Real programmers then test it and improve it. This repeating cycle is Code, Test, Improve. You are the programmer; AI is your helper.

Code	Ask AI for a first version using a clear prompt. Read it line by line and understand it.
Test	Run the program with normal inputs and also edge cases such as 0, negative numbers, empty input and very large numbers.
Improve	If something fails, find the cause, ask AI for help with the exact error, fix and test again.
Record your tests	Write inputs, expected output and actual output in a table. It shows whether the program is correct.



Repeat until the program works correctly

```
def is_even(n):
    return n % 2 == 0

for x in [4, 7, 0, -3]:
    print(x, is_even(x))
```

Real-life example
Builders test a bridge with heavy loads before opening it. Programmers test code with unusual inputs before sharing it. Testing finds problems early.

Lesson at a glance



Discuss with a partner: where do you see 'Code, Test, Improve' in real life? Write your best example.

_____

_____

LESSON 2.5 | PRACTISE: CODE, TEST, IMPROVE

Chapter 2: Communicate with AI

Try it

Task
The program above checks even numbers. Predict the output for 4, 7, 0 and -3. Then write one more edge case you would test.

Check your understanding

1. What are the three steps of the cycle?

2. What is an edge case? Give an example.

3. Why should you read AI-written code line by line?

4. What does n % 2 find?

5. Why record test results in a table?

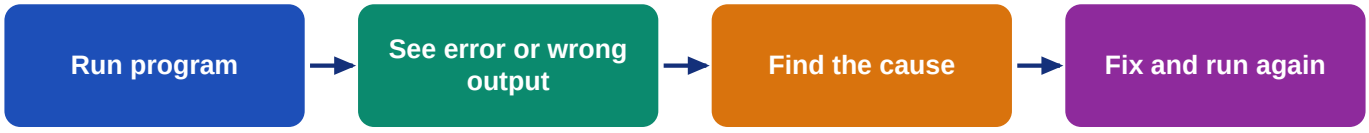
Remember
- Code, Test, Improve is a repeating cycle.
- Test normal inputs and edge cases.
- You must understand code that AI writes.

LESSON 2.6 | DEBUGGING AND RESPONSIBLE CODING

Chapter 2: Communicate with AI - Learn

A bug is a mistake in a program. Debugging means finding and fixing it. Good programmers stay calm, read error messages and use AI as a helper, not as a replacement for thinking.

Read the error	Python tells you the line number and the type of error. Start there.
Common bugs	Wrong spelling of a variable, missing colon or brackets, wrong indentation, and wrong range or condition (logic errors).
Ask AI wisely	Share the code and the exact error. Ask the AI to explain the problem before giving the fix, so that you learn.
Responsible coding	Do not copy code you do not understand. Do not paste private data or passwords. Say when AI helped you.




```
total = 0
for i in range(1, 5):
    total = total + i
print("Sum of 1 to 5 is", total)
```

Real-life example
The program above should print 15 but prints 10. The loop range(1, 5) stops at 4. Changing it to range(1, 6) fixes the logic error.

Lesson at a glance



Discuss with a partner: where do you see 'Debugging and Responsible Coding' in real life? Write your best example.

_____

_____

LESSON 2.6 | PRACTISE: DEBUGGING AND RESPONSIBLE CODING

Chapter 2: Communicate with AI

Try it

Task
Find the bug in the code above. Write the corrected line and a short prompt you could give an AI to explain this bug.

Check your understanding

1. What is debugging?

2. Name two common causes of bugs.

3. What should you share with AI when you get an error?

4. Why should you not copy AI code blindly?

5. What should never be pasted into an AI tool?

Remember
- Read the error message first.
- Understand the fix before using it.
- Protect private data and be honest about AI help.

ACTIVITY 2.1 | PROMPT MAKEOVER

Chapter 2: Communicate with AI - Do

Time: 20 min **Group:** Pairs **Materials:** Worksheet, pencil, AI tool (optional)

Goal

Turn vague prompts into clear 5-part prompts and compare the results.

Steps

1. Take these vague prompts: 'Write code', 'Help me with maths', 'Make a game'.

2. Rewrite each one using Role, Task, Context, Format and Constraints.

3. If an AI tool is available, try both versions and compare.

4. Swap with another pair and score their prompts out of 5.

5. Improve any prompt that scored less than 4.

Record your results

Vague prompt	My improved prompt	Score (out of 5)

Think and write

Which part of RTCFC made the biggest difference to the answer, and why?

_____

_____

_____

_____

ACTIVITY 2.2 | SAME TASK, THREE ROLES

Chapter 2: Communicate with AI - Do

Time: 25 min **Group:** Groups of 3 **Materials:** AI tool or teacher-prepared answers

Goal

See how changing the role in a prompt changes the answer.

Steps

1. Choose one topic, for example 'Why is exercise important?'.

2. Write the same task with 3 roles: a doctor, a sports coach, a Class 3 teacher.

3. Run each prompt and read the three answers.

4. Compare style, vocabulary and level of difficulty.

5. Choose which role is best for a Class 3 student and explain why.

Record your results

Role	Style of the answer	Best for whom?

Think and write

When would you choose each role in real life? Give one example.

_____

_____

_____

_____

ACTIVITY 2.3 | CODE, TEST, IMPROVE LAB

Chapter 2: Communicate with AI - Do

Time: 35 min **Group:** Individual **Materials:** Computer with Python or an online Python editor

Goal

Practise the Code, Test, Improve cycle on a small program.

Steps

1. Ask AI: 'Write a Python program to check whether a number is even or odd. Add comments.'

2. Read the code and explain each line in your notebook.

3. Run it with 4, 7, 0 and -3 and record the results.

4. Ask AI to handle wrong input such as letters.

5. Test again and record what changed.

Record your results

Input	Expected output	Actual output

Think and write

Which test was the most useful? Did the program have a bug before improvement?

_____

_____

_____

_____

ACTIVITY 2.4 | BUG HUNT RELAY

Chapter 2: Communicate with AI - Do

Time: 20 min **Group:** Teams of 4 **Materials:** Five short programs with one bug each (teacher gives)

Goal

Find bugs by reading and tracing code by hand.

Steps

1. Each team gets 5 short programs on paper.

2. Read each program line by line and write what it should do.

3. Trace the output by hand using a small table.

4. Mark the bug and write the corrected line.

5. Present one bug and its fix to the class.

Record your results

Program no.	Bug found	Corrected line

Think and write

Which kind of bug was hardest to find: spelling, brackets, indentation or logic?

_____

_____

_____

_____

ACTIVITY 2.5 | BUILD A PROMPT LIBRARY

Chapter 2: Communicate with AI - Do

Time: 30 min **Group:** Groups of 4 **Materials:** Chart paper, markers, sticky notes

Goal

Create reusable prompts for studying and coding that your class can share.

Steps

1. Brainstorm 5 tasks you do often (summarise notes, make a quiz, explain a concept, debug code, plan revision).

2. Write one 5-part prompt for each task.

3. Test two prompts with an AI tool or with a peer acting as AI.

4. Improve the prompts based on the results.

5. Display the library on a chart for the class.

Record your results

Task	My final prompt	How well did it work?

Think and write

Which prompt will you use most often? Why?

_____

_____

_____

_____

MINI PROJECT | PART 1: PLAN

Chapter 2: Communicate with AI

My Prompt Toolkit and Mini Program

Create a toolkit of 5 tested prompts for school work and write one small Python program with AI help. Show the Code, Test, Improve steps you followed.

Pick an idea (tick one)

[ ] Marks average and grade calculator
[ ] Simple quiz game with 5 questions
[ ] Multiplication table generator
[ ] Daily study planner that prints a timetable

Project plan

Plan	My answer
Project name	
Problem I am solving	
Who will use it	
Tools / AI I will use	
Data or materials needed	
Team members and roles	
Time I need	

Success criteria

- Has 5 clear prompts that follow RTCFC.
- Python program runs and has comments.
- Test table with at least 4 tests including 1 edge case.
- Explains what changed after testing and improvement.

MINI PROJECT | PART 2: BUILD, TEST, PRESENT

Chapter 2: Communicate with AI

Build tip

Write your own comments in the code. Run every version. Keep a record of AI prompts you used. Never include real names, phone numbers or passwords in test data.

Test log

Test	What I tried	Result	What I will improve
1			
2			
3			
4			

Presentation checklist (tick when done)

[ ] I can explain the problem in one sentence.
[ ] I can show how my project works.
[ ] I can say where AI or data is used.
[ ] I can name one limit or risk.
[ ] I checked facts and privacy.
[ ] I thanked my team and mentor.

Peer feedback: 2 stars and a wish

Star 1	Star 2	Wish (one improvement)

Project score (teacher)

Idea and purpose	Working prototype	Responsible AI use	Presentation	Total
5	5	5	5	/ 20

WORKSHEET A | CHAPTER 2: COMMUNICATE WITH AI

Name: _____ Class: _____ Date: _____ Marks: ____ / 18

Section A - Choose the correct answer (8 x 1 = 8)

1. A good prompt should include:
   (a) only one word
   (b) a clear task, context and format
   (c) only emojis
   (d) no details

2. In RTCFC, the letter F stands for:
   (a) Fast
   (b) Format
   (c) Fiction
   (d) Folder

3. In 'Code, Test, Improve', Test means:
   (a) admiring the code
   (b) running with different inputs including edge cases
   (c) deleting the code
   (d) printing the code

4. If AI-written code shows an error, the best step is to:
   (a) ignore it
   (b) read the error, understand it, fix and re-test
   (c) copy random code
   (d) switch off the computer

5. Which Python line displays Hello?
   (a) echo("Hello")
   (b) say Hello
   (c) print("Hello")
   (d) display Hello

6. Which symbol gives the remainder in Python?
   (a) /
   (b) %
   (c) #
   (d) &

7. Which prompt is the best?
   (a) Write code
   (b) Act as a Python teacher. Write a program to check even or odd. Add comments.
   (c) Python
   (d) Do it fast

8. A good edge case to test a division program is dividing by:
   (a) 2
   (b) 0 (zero)
   (c) 10
   (d) 5

Section B - Fill in the blanks (6 x 1 = 6)

1. Telling AI to 'act as a Python teacher' gives it a _____.

2. Testing with unusual inputs like 0 or an empty list means checking _____ cases.

3. Finding and fixing errors in a program is called _____.

4. A named box that stores a value in a program is a _____.

5. The Python function that takes input from a user is _____().

6. Asking follow-up prompts to improve an answer is called _____ the prompt.

Section C - True or False (4 x 1 = 4)

1. AI-generated code is always correct. **(True / False)**

2. You should run and test code that AI gives you. **(True / False)**

3. Adding context to a prompt gives the AI better information. **(True / False)**

4. In Python, print() is used to take input from the user. **(True / False)**

WORKSHEET B | CHAPTER 2: COMMUNICATE WITH AI

Name: _____ Class: _____ Date: _____ Marks: ____ / 24

Section D - Match the columns (5 x 1 = 5)

Column A	Answer	Column B
1. Role	_____	(a) How the answer should look
2. Context	_____	(b) A mistake in a program
3. Format	_____	(c) Who the AI should act as
4. Constraint	_____	(d) A limit such as word count
5. Bug	_____	(e) Background information for the task

Section E - Short answers (4 x 3 = 12)

1. Rewrite the vague prompt 'Make a program' as a clear prompt.

_____

_____

_____

_____

2. Why should you not copy AI-generated code without understanding it?

_____

_____

_____

_____

3. What is an edge case? Give one example.

_____

_____

_____

_____

4. Explain the Code, Test, Improve cycle in three lines.

_____

_____

_____

_____

Section F - Find the bug (code below) (7)

This program should print the sum of 1 to 5 (which is 15) but prints 10. Find the bug, write the corrected line and write a better prompt you could give AI.

```
total = 0
for i in range(1, 5):
    total = total + i
print("Sum of 1 to 5 is", total)
```

_____

_____

_____

_____

_____

_____

_____

_____

_____

REVISION AND PUZZLE

Chapter 2: Communicate with AI

This chapter in 10 points

1. A prompt is the instruction you give to AI.

2. Clear prompts give better answers than vague prompts.

3. RTCFC: Role, Task, Context, Format, Constraints.

4. Examples, step-by-step requests and follow-ups improve results.

5. Always verify AI answers.

6. print() shows output and input() takes input in Python.

7. Variables store values; if/else decide; loops repeat.

8. Code, Test, Improve is a repeating cycle.

9. Test normal inputs and edge cases and record the results.

10. Read errors, understand fixes, protect private data and disclose AI help.

Word search: find these 8 words (across, down or diagonal)

O	J	P	O	N	N	U	D	I	P	M	H
O	U	P	C	C	T	E	P	Y	I	T	O
A	Y	M	N	O	A	L	Y	T	L	P	I
L	J	R	G	N	B	V	S	P	I	D	M
D	R	B	G	T	J	A	D	E	B	U	G
A	O	T	R	E	I	M	P	R	O	V	E
J	L	R	L	X	I	P	Y	T	H	O	N
D	E	B	E	T	G	A	H	I	R	I	F
H	G	U	W	K	J	T	E	S	T	L	B
S	U	F	R	V	F	O	R	M	A	T	G
L	P	R	O	M	P	T	P	M	H	M	S
S	K	P	U	B	H	V	N	P	K	C	I

PROMPT | ROLE | CONTEXT | FORMAT | PYTHON | DEBUG | TEST | IMPROVE

GLOSSARY AND SELF-CHECK

Chapter 2: Communicate with AI

Glossary

Prompt	The instruction you give to an AI tool.
Role	Who you want the AI to act as.
Context	Background that helps the AI understand your need.
Format	How you want the answer to look, such as a table or list.
Constraint	A limit, such as 'under 50 words'.
Edge case	An unusual input such as 0, negative numbers or empty data.
Bug	A mistake in a program.
Debugging	Finding and fixing bugs.
Iteration	Repeating and improving step by step.
Variable	A named box in a program that stores a value.

I can ... (tick the box)

☐	Explain what a prompt is.	Yes / Need help
☐	Write a prompt with Role, Task, Context, Format and Constraints.	Yes / Need help
☐	Improve a prompt using follow-ups.	Yes / Need help
☐	Write a simple Python program with print, input and if.	Yes / Need help
☐	Explain the Code, Test, Improve cycle.	Yes / Need help
☐	Create test cases including edge cases.	Yes / Need help
☐	Debug a simple error.	Yes / Need help
☐	Use AI for coding responsibly.	Yes / Need help

My reflection

The most interesting thing I learned in this chapter was ...

_____

_____

_____

Teacher's remarks: _____ Signature: _____ Date: _____

CHAPTER 3 | MONTH 3 | AI APPLICATIONS

AI at Work | AI Solving Real Problems



In this chapter you will

- Identify how AI is used in agriculture, health, education, business and transport.
- Use a five-step canvas to plan an AI solution for a real problem.
- Weigh the benefits and risks of AI and explain why humans stay responsible.

Key words

Automation	Machines doing repeated tasks with little human effort.
Prediction	An AI's estimate of what is likely to happen.
Chatbot	A program that talks with people using text or voice.
Fraud detection	Spotting unusual or false transactions.
Image recognition	AI identifying objects or problems in pictures.
Sensor	A device that measures things like heat, moisture or motion.
Benefit	A good result for people.

Think about it
Which job in your town could be made easier by AI? Which part must always stay with a human?

WARM-UP | THINK BEFORE YOU LEARN

Chapter 3: AI Applications

Before you start this chapter, find out what you already know. There are no wrong answers here. Guess, think and write.

My KWL chart

K - What I already KNOW	W - What I WANT to know	L - What I LEARNED (fill at the end)

Guess and write

1. Name one job or place where you have seen a machine do work that people used to do.

_____

_____

2. How do you think a navigation app knows about traffic?

_____

_____

3. What problem in your school or town would you like to solve with technology?

_____

_____

4. Do you think AI will help workers or replace them? Why?

_____

_____

5. Write one question you want answered in this chapter.

_____

_____

Chapter roadmap

Lesson 3.1
AI at Work: Automation and Assistance

Lesson 3.2
AI in Agriculture and Environment

Lesson 3.3
AI in Health and Education

Lesson 3.4
AI in Business, Banking and Transport

Lesson 3.5
Solving Real Problems with AI

Lesson 3.6
Benefits, Risks and the Human Role

LESSON 3.1 | AI AT WORK: AUTOMATION AND ASSISTANCE

Chapter 3: AI Applications - Learn

AI helps people in workplaces in two main ways. It automates repeated tasks, and it assists people by finding patterns and giving suggestions. In most jobs, people stay responsible for the final decision.

Automation	AI does repetitive work such as sorting emails, reading forms or checking products on a belt.
Assistance	AI gives suggestions, like highlighting a risky loan or suggesting a reply, while a human decides.
Speed and scale	AI can check thousands of items quickly, which would take people a very long time.
New jobs	AI also creates new jobs such as data labellers, AI trainers, AI safety checkers and prompt designers.

Automation	Assistance
• AI does the task • Repeated, rule-based work • Example: sorting parcels • Human supervises	• AI suggests, human decides • Needs judgement • Example: doctor reads AI scan hint • Human is responsible

Real-life example
In an office, AI can write the first draft of a meeting summary. The employee reads it, fixes errors and sends it. The AI saves time, but the person is accountable.

Lesson at a glance



Discuss with a partner: where do you see 'AI at Work: Automation and Assistance' in real life? Write your best example.

_____

_____

LESSON 3.1 | PRACTISE: AI AT WORK: AUTOMATION AND ASSISTANCE

Chapter 3: AI Applications

Try it

Task
Think of any workplace (shop, hospital, school, bank, farm). List 3 tasks AI could automate and 2 tasks that need a human.

Check your understanding

1. What is automation?

2. How is AI assistance different from automation?

3. Why should humans stay responsible for final decisions?

4. Name two new jobs created by AI.

5. Give one example of AI at a workplace you know.

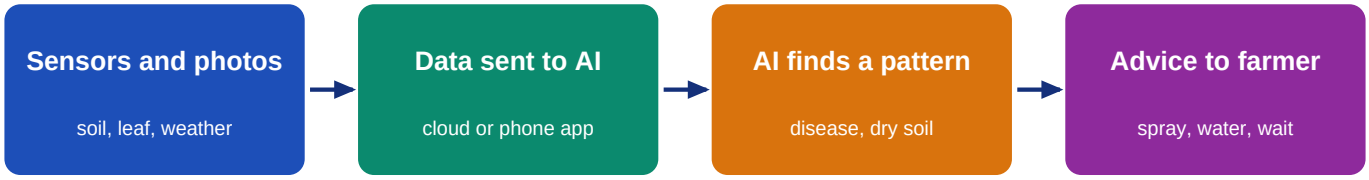
Remember
- AI automates tasks and assists people.
- Humans remain responsible for decisions.
- AI creates new kinds of jobs.

LESSON 3.2 | AI IN AGRICULTURE AND ENVIRONMENT

Chapter 3: AI Applications - Learn

Farming and the environment depend on weather, soil, water and pests. AI helps by reading data from cameras, sensors and satellites to give early warnings and better advice.

Crop disease detection	A phone camera photo of a leaf can be checked by an AI model to suggest if the plant looks healthy or diseased.
Weather and rainfall	AI models study weather data to help forecast rain, heat and storms so farmers can plan.
Smart irrigation	Soil moisture sensors with AI decide when and how much to water, saving water.
Environment	AI helps monitor air quality, sort waste, track forests and protect wildlife with camera-trap images.



Real-life example
A farmer photographs a spotted leaf. The app compares it with thousands of labelled leaf photos and suggests a likely disease and a nearby expert to confirm.

Lesson at a glance



Discuss with a partner: where do you see 'AI in Agriculture and Environment' in real life? Write your best example.

_____

_____

LESSON 3.2 | PRACTISE: AI IN AGRICULTURE AND ENVIRONMENT

Chapter 3: AI Applications

Try it

Task
Pick a crop grown in your area. List 3 problems the farmer faces and write how AI data (photo, sensor, weather) could help with each.

_____

_____

_____

_____

_____

Check your understanding

1. How can AI help detect a plant disease?

_____

_____

2. What is smart irrigation?

_____

_____

3. Why is weather forecasting important for farmers?

_____

_____

4. Name one way AI helps protect the environment.

_____

_____

5. Why should a farmer confirm AI advice with an expert?

_____

_____

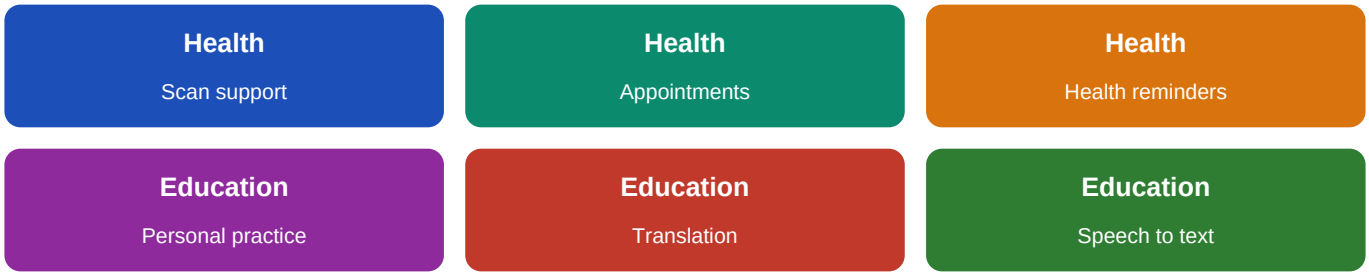
Remember
- AI uses photos, sensors and weather data in farming.
- It gives early warnings and saves water.
- Experts should confirm important decisions.

LESSON 3.3 | AI IN HEALTH AND EDUCATION

Chapter 3: AI Applications - Learn

In health, AI supports doctors and nurses. In education, AI helps students learn at their own pace. In both fields, privacy and accuracy matter very much.

Medical images	AI can help doctors by highlighting suspicious areas in scans or X-rays. The doctor makes the final decision.
Hospital support	AI helps with appointment scheduling, record keeping and reminders to take medicines.
Personalised learning	Learning apps notice which topics you find hard and give extra practice.
Accessibility	Speech-to-text, text-to-speech and live translation help students with different needs and languages.



Real-life example
A learning app sees that a student makes mistakes in fractions. It gives 5 more fraction questions with hints and then moves to the next topic when the student improves.

Lesson at a glance



Discuss with a partner: where do you see 'AI in Health and Education' in real life? Write your best example.

_____

_____

LESSON 3.3 | PRACTISE: AI IN HEALTH AND EDUCATION

Chapter 3: AI Applications

Try it

Task
Imagine an AI study helper for your class. Write 3 features it should have and 2 things it should never do (for example, share your marks).

Check your understanding

1. How can AI support a doctor?

2. Why must a doctor make the final decision?

3. What is personalised learning?

4. How does AI help students who speak different languages?

5. Why is privacy important in health and education?

Remember
- AI supports doctors, nurses and teachers.
- Personalised learning adapts to each student.
- Privacy and accuracy are very important.

LESSON 3.4 | AI IN BUSINESS, BANKING AND TRANSPORT

Chapter 3: AI Applications - Learn

Companies, banks and transport systems use AI to save time and serve people better. These systems handle huge amounts of data every day.

Banking	AI spots unusual transactions to prevent fraud and answers customer questions through chatbots.
Business	Shops use AI to forecast demand, manage stock and recommend products.
Navigation	Map apps use data from many phones to estimate traffic and suggest faster routes.
Smart cities	AI can control traffic signals, plan bus routes and manage waste collection and street lights.

Area	AI use	Benefit
Banking	Fraud detection	Safer money
Business	Demand forecasting	Less waste, right stock
Transport	Traffic prediction	Faster, safer travel
City	Smart signals	Less jam and pollution

Real-life example
A bank's AI notices a card used in two far cities within an hour. It pauses the payment and sends the customer a message to confirm.

Lesson at a glance



Discuss with a partner: where do you see 'AI in Business, Banking and Transport' in real life? Write your best example.

_____

_____

LESSON 3.4 | PRACTISE: AI IN BUSINESS, BANKING AND TRANSPORT

Chapter 3: AI Applications

Try it

Task
Choose a shop or bus route you know. Write how AI data (sales, location, time) could improve it. Name the data needed.

Check your understanding

1. How does AI help prevent bank fraud?

2. What does a demand forecast do for a shop?

3. How do navigation apps know about traffic?

4. Give one smart city use of AI.

5. Why might a customer still want to talk to a human at a bank?

Remember
- AI detects fraud and answers customer queries.
- AI forecasts demand and traffic.
- Humans are still needed for complex or sensitive cases.

LESSON 3.5 | SOLVING REAL PROBLEMS WITH AI

Chapter 3: AI Applications - Learn

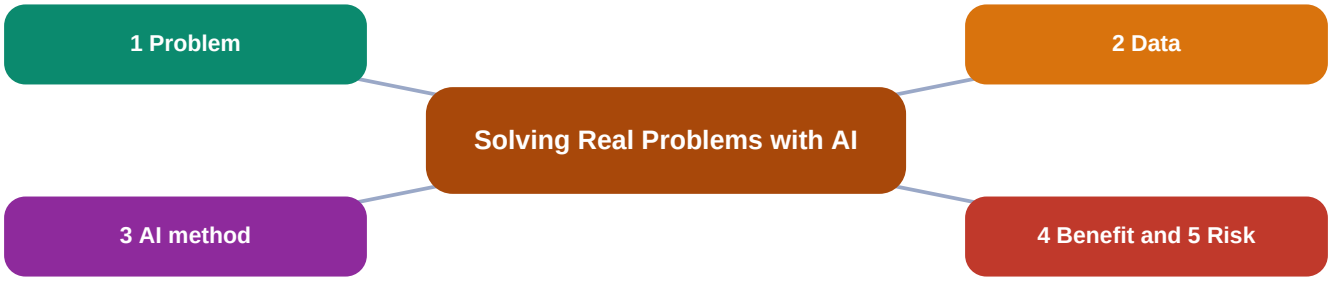
AI is useful only when it solves a real problem. Use a simple five-step canvas to plan: Problem, Data, AI method, Benefit and Risk. Start small and test with real people.

1 Problem	Describe exactly what is wrong and who is affected. Be specific.
2 Data	What information is needed? Photos, numbers, text or sensor readings? Do you have permission to use it?
3 AI method	Prediction, image recognition, chatbot or recommendation: pick the simplest method that works.
4 Benefit and 5 Risk	Who gains and how much? What could go wrong: errors, privacy, bias, cost, job impact?



Real-life example
Problem: students waste canteen food. Data: past daily sales and attendance. Method: prediction. Benefit: less waste, saved money. Risk: wrong estimates on rainy days, so keep a small buffer.

Lesson at a glance



Discuss with a partner: where do you see 'Solving Real Problems with AI' in real life? Write your best example.

_____

_____

LESSON 3.5 | PRACTISE: SOLVING REAL PROBLEMS WITH AI

Chapter 3: AI Applications

Try it

Task
Choose a problem in your school or street. Fill all five steps of the canvas in a short sentence each.

Check your understanding

1. Why must a problem be defined clearly?

2. What kind of data might a canteen problem need?

3. Name two AI methods.

4. Give one benefit and one risk of an AI solution you thought of.

5. Why should you test with real people?

Remember
- Canvas: Problem, Data, AI method, Benefit, Risk.
- Choose the simplest method that works.
- Test with real users and watch for risks.

LESSON 3.6 | BENEFITS, RISKS AND THE HUMAN ROLE

Chapter 3: AI Applications - Learn

Every technology has benefits and risks. Responsible AI users ask: who benefits, who might be harmed and who is in charge? A person who checks and approves AI decisions is called a 'human in the loop'.

Benefits	Saves time, reduces errors in repeated tasks, works day and night, and can reach remote places.
Risks	Wrong results, unfair decisions from biased data, privacy leaks, over-dependence and job changes.
Human in the loop	Important decisions on health, money, law or safety must have a human who checks the AI result.
Stakeholders	Think about everyone affected: users, workers, customers and the community.

Benefits	Risks
• Saves time • Fewer repeated errors • Available 24 x 7 • Reaches remote areas	• Wrong or unfair results • Privacy leaks • Over-dependence • Job changes

Real-life example
A loan app uses AI to rank applicants. If its data is biased against some areas, good applicants may be rejected. A human review and regular fairness checks reduce this risk.

Lesson at a glance



Discuss with a partner: where do you see 'Benefits, Risks and the Human Role' in real life? Write your best example.

_____

_____

LESSON 3.6 | PRACTISE: BENEFITS, RISKS AND THE HUMAN ROLE

Chapter 3: AI Applications

Try it

Task
Pick one AI application from this chapter. Write 2 stakeholders, 1 benefit and 1 risk for each stakeholder.

Check your understanding

1. What is a human in the loop?

2. Why do important decisions need human checking?

3. Name two risks of AI.

4. Who are stakeholders? Give two examples.

5. How can biased data affect a loan decision?

Remember
- Weigh benefits and risks every time.
- Keep a human in the loop for important decisions.
- Think about all stakeholders.

ACTIVITY 3.1 | INDUSTRY MATCH-UP

Chapter 3: AI Applications - Do

Time: 15 min **Group:** Pairs **Materials:** 12 cards: 6 industries and 6 AI uses (teacher or students make them)

Goal

Connect industries with the AI use that helps them most.

Steps

1. Lay all industry cards on one side and AI-use cards on the other.

2. Match each industry to the best AI use.

3. Add one more AI use for each industry from your own idea.

4. Compare with another pair and discuss any differences.

5. Write the final matches in the table.

Record your results

Industry	AI use	My own extra idea

Think and write

Which match was hardest, and why?

_____

_____

_____

_____

ACTIVITY 3.2 | AI PROBLEM-SOLVER CANVAS

Chapter 3: AI Applications - Do

Time: 40 min **Group:** Groups of 4 **Materials:** Chart paper, markers

Goal

Plan an AI solution for a real local problem using the five-step canvas.

Steps

1. Choose a real problem from school, home or town (waste, traffic, water use, library books).

2. Write the problem in one clear sentence.

3. List the data needed and where it comes from.

4. Choose an AI method and write expected benefit and risk.

5. Prepare a 2-minute pitch and present to the class.

Record your results

Canvas step	Our answer	Questions from classmates

Think and write

What feedback did you get? How will you improve your plan?

_____

_____

_____

_____

ACTIVITY 3.3 | DEBATE: AI AT WORK

Chapter 3: AI Applications - Do

Time: 25 min **Group:** Two teams **Materials:** Notebook, timer

Goal

Practise listening, evidence and respectful disagreement about AI and jobs.

Steps

1. Motion: 'AI will help workers more than it will replace them'.

2. Split into For and Against teams.

3. Prepare 3 points with examples (10 minutes).

4. Take turns to speak for 1 minute each.

5. Vote again and note if anyone changed their mind.

Record your results

Point	For / Against	Example used

Think and write

Which argument from the other team was strongest? Why?

_____

_____

_____

_____

ACTIVITY 3.4 | AI AROUND MY TOWN: SURVEY

Chapter 3: AI Applications - Do

Time: 30 min (plus home task) **Group:** Individual then groups **Materials:** Survey sheet, pencil

Goal

Find real examples of AI in your surroundings by observing and asking people.

Steps

1. Ask 3 adults: 'Which app or machine with AI or smart features do you use in your work or daily life?'.

2. Note what the tool does and how it helps.

3. Ask: 'Has it ever made a mistake? What did you do?'.

4. Bring the data to class and combine in groups.

5. Find the most common AI uses in your community.

Record your results

Person (job)	AI or smart tool	Benefit / mistake seen

Think and write

What surprised you? Which job uses AI in a way you did not expect?

_____

_____

_____

_____

ACTIVITY 3.5 | STAKEHOLDER ROLE PLAY

Chapter 3: AI Applications - Do

Time: 30 min **Group:** Groups of 5 **Materials:** Role cards: farmer, app developer, bank manager, student, community leader

Goal

See an AI solution from different points of view.

Steps

1. Choose an AI solution, for example 'AI app that predicts crop disease'.

2. Each student takes one role card.

3. Each person says what they like and what they worry about.

4. The group lists common benefits and common worries.

5. Suggest changes to make the solution fairer and safer.

Record your results

Role	What I like	What I worry about

Think and write

How did changing roles change your opinion?

_____

_____

_____

_____

MINI PROJECT | PART 1: PLAN

Chapter 3: AI Applications

AI Solution Pitch for a Local Problem

Choose a real problem near you and design an AI solution on paper. Present your canvas, a simple model of the idea (drawing or flowchart) and a risk plan. Optionally create a prototype using a no-code tool.

Pick an idea (tick one)

[ ] Reduce food waste in the canteen
[ ] Predict crowded bus times at your stop
[ ] Detect plant leaf problems in a school garden
[ ] Smart reminders to save water or electricity

Project plan

Plan	My answer
Project name	
Problem I am solving	
Who will use it	
Tools / AI I will use	
Data or materials needed	
Team members and roles	
Time I need	

Success criteria

- Problem is clear and real.
- Data needs and privacy are explained.
- Benefits and risks are balanced.
- Human role in the solution is described.

MINI PROJECT | PART 2: BUILD, TEST, PRESENT

Chapter 3: AI Applications

Build tip

Talk to the people who face the problem before designing. Draw a flow from data to AI to decision to human check. Keep the idea small enough to test in a week.

Test log

Test	What I tried	Result	What I will improve
1			
2			
3			
4			

Presentation checklist (tick when done)

[ ] I can explain the problem in one sentence.
[ ] I can show how my project works.
[ ] I can say where AI or data is used.
[ ] I can name one limit or risk.
[ ] I checked facts and privacy.
[ ] I thanked my team and mentor.

Peer feedback: 2 stars and a wish

Star 1	Star 2	Wish (one improvement)

Project score (teacher)

Idea and purpose	Working prototype	Responsible AI use	Presentation	Total
5	5	5	5	/ 20

WORKSHEET A | CHAPTER 3: AI APPLICATIONS

Name: _____ Class: _____ Date: _____ Marks: ____ / 18

Section A - Choose the correct answer (8 x 1 = 8)

1. AI that detects disease from photos of crop leaves is used in:
   (a) sports
   (b) agriculture
   (c) music
   (d) cooking

2. Banks mostly use AI to:
   (a) paint buildings
   (b) spot unusual or fraudulent transactions
   (c) print notes
   (d) cut keys

3. The FIRST step in solving a problem with AI is to:
   (a) choose a fancy tool
   (b) clearly define the problem
   (c) publish the app
   (d) buy a computer

4. A chatbot answering customer questions all day is AI in:
   (a) farming
   (b) space travel
   (c) customer service
   (d) mining

5. Smart irrigation mainly saves:
   (a) water
   (b) paper
   (c) internet
   (d) time on tests

6. A 'human in the loop' means:
   (a) a robot repeating work
   (b) a person who checks AI decisions
   (c) a loop in code
   (d) a human replaced by AI

7. Navigation apps estimate traffic using:
   (a) guessing
   (b) data from many phones
   (c) television
   (d) newspapers

8. Which is a risk of AI?
   (a) Faster service
   (b) Privacy leaks
   (c) Less paperwork
   (d) Reaching remote areas

Section B - Fill in the blanks (6 x 1 = 6)

1. Navigation apps use AI to predict _____ and suggest faster routes.

2. Factory cameras with AI check product _____.

3. AI needs _____ to learn about a problem.

4. The five-step canvas ends with Benefit and _____.

5. A person affected by an AI decision is called a _____.

6. Machines doing repeated tasks with little human effort is called _____.

Section C - True or False (4 x 1 = 4)

1. AI works without any data. **(True / False)**

2. Doctors should make the final medical decision, not AI. **(True / False)**

3. All AI results are always fair. **(True / False)**

4. Personalised learning adapts to each student. **(True / False)**

WORKSHEET B | CHAPTER 3: AI APPLICATIONS

Name: _____ Class: _____ Date: _____ Marks: ____ / 24

Section D - Match the columns (5 x 1 = 5)

Column A	Answer	Column B
1. Fraud detection	_____	(a) Device that measures moisture, heat or motion
2. Chatbot	_____	(b) Estimate of what is likely to happen
3. Sensor	_____	(c) Spotting unusual or false transactions
4. Stakeholder	_____	(d) Person or group affected by a decision
5. Prediction	_____	(e) Program that talks with people

Section E - Short answers (4 x 3 = 12)

1. Name one problem in your school or town. How can AI help solve it?

2. Give one risk of using AI in healthcare and how to reduce it.

3. Explain the five steps of the AI Problem-Solving Canvas.

4. Why should humans stay in the loop for important decisions?

Section F - Case study canvas (7)

A farmer is losing part of his crop to pests and does not know which treatment to use or when. Fill the AI Problem-Solving Canvas below in short points.

REVISION AND PUZZLE

Chapter 3: AI Applications

This chapter in 10 points

1. AI automates repeated tasks and assists human decisions.

2. Humans stay responsible for final decisions.

3. In farming AI uses photos, sensors and weather data.

4. AI supports doctors, teachers and students with privacy care.

5. Banks use AI for fraud detection and chatbots.

6. Navigation and smart city tools use data to reduce traffic problems.

7. Solve real problems with the canvas: Problem, Data, AI method, Benefit, Risk.

8. Choose the simplest method and test with real users.

9. Every AI use has benefits and risks.

10. Keep a human in the loop and think about all stakeholders.

Word search: find these 8 words (across, down or diagonal)

C	O	N	P	B	J	B	I	W	J	B	N
H	F	B	E	N	E	F	I	T	M	C	K
V	S	E	N	S	O	R	E	D	A	T	A
C	E	P	B	I	E	V	A	R	P	A	G
L	H	E	L	U	N	P	U	T	P	J	E
P	S	A	T	I	E	A	T	M	W	E	Y
A	R	I	T	T	K	V	O	H	T	N	I
W	M	E	E	B	E	E	M	R	D	M	D
F	A	V	D	O	O	E	A	I	C	D	V
J	E	J	S	I	K	T	T	S	M	P	H
F	R	A	U	D	C	U	E	K	F	F	F
M	G	Y	S	B	U	T	Y	R	B	A	M

SENSOR | FRAUD | CHATBOT | PREDICT | AUTOMATE | RISK | DATA | BENEFIT

GLOSSARY AND SELF-CHECK

Chapter 3: AI Applications

Glossary

Automation	Machines doing repeated tasks with little human effort.
Prediction	An AI's estimate of what is likely to happen.
Chatbot	A program that talks with people using text or voice.
Fraud detection	Spotting unusual or false transactions.
Image recognition	AI identifying objects or problems in pictures.
Sensor	A device that measures things like heat, moisture or motion.
Benefit	A good result for people.
Risk	A possible harm or problem.
Stakeholder	A person or group affected by a decision.
Human in the loop	A person who checks and approves AI decisions.

I can ... (tick the box)

☐	Name AI uses in farming, health, education and banking.	Yes / Need help
☐	Explain automation and assistance.	Yes / Need help
☐	Use the five-step canvas.	Yes / Need help
☐	Describe benefits and risks of an AI tool.	Yes / Need help
☐	Explain 'human in the loop'.	Yes / Need help
☐	Identify stakeholders of an AI solution.	Yes / Need help
☐	Pitch a solution to a real problem.	Yes / Need help
☐	Say why privacy matters in AI applications.	Yes / Need help

My reflection

The most interesting thing I learned in this chapter was ...

_____

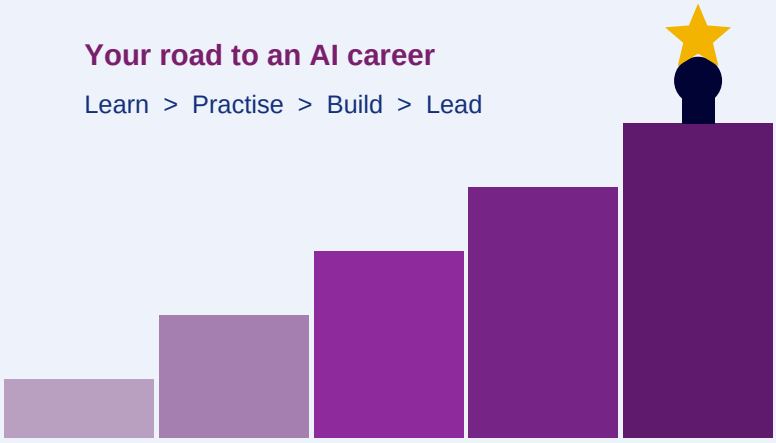
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Teacher's remarks: _____ Signature: _____ Date: _____

CHAPTER 4 | MONTH 4 | AI & CAREER

Your Road to an AI Career | AI Skills Beyond School



In this chapter you will

- Know the different careers connected to AI and what each person does.
- Identify the technical and human skills needed and plan your own roadmap.
- Learn how to build skills beyond the classroom through projects, courses and a portfolio.

Key words

Data Analyst	Finds insights in data.
Data Scientist	Builds models and studies data to answer questions.
ML Engineer	Builds and runs machine learning systems.
Domain knowledge	Deep understanding of a field like health or farming.
Statistics	Maths of collecting and understanding data.
Portfolio	A collection of your projects.
Internship	Short work experience to learn skills.

Did you know?
Ada Lovelace wrote notes in the 1840s that include what is often called the first computer program.

WARM-UP | THINK BEFORE YOU LEARN

Chapter 4: AI & Career

Before you start this chapter, find out what you already know. There are no wrong answers here. Guess, think and write.

My KWL chart

K - What I already KNOW	W - What I WANT to know	L - What I LEARNED (fill at the end)

Guess and write

1. What do you want to become? Write 2 careers you are curious about.

_____

_____

2. Which of your school subjects do you enjoy most? Why?

_____

_____

3. How do you think AI will change the job you are thinking about?

_____

_____

4. Name one skill you are good at and one you want to improve.

_____

_____

5. Who inspires you in the technology world?

_____

_____

Chapter roadmap

Lesson 4.1
The World of AI Careers

Lesson 4.2
AI Plus Every Field

Lesson 4.3
Skills for the AI Age

Lesson 4.4
Your Path After Class 10

Lesson 4.5
Building Skills Beyond School

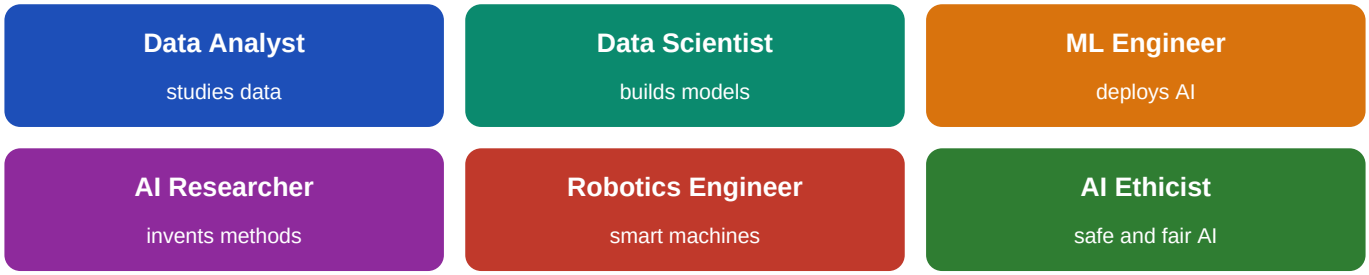
Lesson 4.6
Learning for Life: Future of Work

LESSON 4.1 | THE WORLD OF AI CAREERS

Chapter 4: AI & Career - Learn

AI is not one job. It is a whole family of careers. Some people build AI, some use it, some check it, and some make rules for it. Let us meet the main roles.

Data Analyst	Collects and studies data to find useful insights and shows them in charts.
Data Scientist and ML Engineer	The data scientist builds models from data. The machine learning engineer makes models work reliably inside real apps.
AI Researcher	Invents new methods and tests new ideas, usually after advanced studies.
Robotics and AI Ethics	Robotics engineers build smart machines. AI ethicists and policy experts make sure AI is safe, fair and legal.



Real-life example
A hospital team has a data analyst who studies patient records, an ML engineer who runs a scan-checking tool, a doctor who makes decisions and an ethics officer who protects privacy.

Lesson at a glance



Discuss with a partner: where do you see 'The World of AI Careers' in real life? Write your best example.

_____

_____

LESSON 4.1 | PRACTISE: THE WORLD OF AI CAREERS

Chapter 4: AI & Career

Try it

Task
Choose two careers from this lesson. For each, write what the person does in a day and which school subject helps most.

Check your understanding

1. Name four AI-related careers.

2. What does a data analyst do?

3. How is an ML engineer different from a data scientist?

4. What is the job of an AI ethicist?

5. Which role sounds most interesting to you and why?

Remember
- AI has many kinds of careers.
- Some build AI, some use it, some guide it.
- Teams with different skills work together.

LESSON 4.2 | AI PLUS EVERY FIELD

Chapter 4: AI & Career - Learn

You do not have to become a programmer to work with AI. Doctors, farmers, lawyers, designers, journalists and sports coaches now use AI as a tool. People who know a field well AND understand AI are very valuable.

Domain knowledge	Deep knowledge of a field, such as farming or medicine, helps you ask the right questions and spot wrong AI answers.
AI as a tool	A designer uses AI for idea sketches. A teacher uses AI to prepare practice questions. A farmer uses AI for weather advice.
Hybrid careers	Examples: AI in healthcare, agri-tech, sports analytics, digital marketing, journalism and legal research.
Your advantage	Your own interests and local knowledge can help you build solutions that outsiders may miss.

Field	How AI helps	Skill to add
Medicine	Scan support, record summaries	Biology + data basics
Agriculture	Crop and weather advice	Farming + sensors
Design	Ideas and drafts	Creativity + prompting
Sports	Performance analysis	Statistics

Real-life example
A cricket coach who understands the game and basic data can use video analysis to find where a batter loses wickets. The AI alone does not know the game; the coach does.

Lesson at a glance



Discuss with a partner: where do you see 'AI Plus Every Field' in real life? Write your best example.

_____

_____

LESSON 4.2 | PRACTISE: AI PLUS EVERY FIELD

Chapter 4: AI & Career

Try it

Task
Pick a field you love (music, farming, law, cooking, sports). Write 3 ways AI could help in that field and what you would still do yourself.

Check your understanding

1. What is domain knowledge?

2. Why is domain knowledge useful when using AI?

3. Give two examples of hybrid careers.

4. Can a farmer use AI? How?

5. How can your own interests become an advantage?

Remember
- AI is a tool in almost every field.
- Domain knowledge plus AI skills is powerful.
- Your interests can guide your AI career.

LESSON 4.3 | SKILLS FOR THE AI AGE

Chapter 4: AI & Career - Learn

Careers in AI need two kinds of skills: technical skills and human skills. Both can be built step by step, starting from Class 10.

Technical skills	Mathematics and statistics, Python programming, working with data, and understanding how AI models work.
Thinking skills	Problem solving, logic, curiosity, asking good questions and checking facts.
Human skills	Communication, teamwork, creativity, leadership and empathy. AI cannot replace these.
Values and habits	Honesty, responsibility, fairness and a habit of learning something new regularly.

Technical skills	Human skills
• Maths and statistics • Python programming • Data handling • Understanding AI models	• Communication • Teamwork • Creativity • Ethics and responsibility

Real-life example
Two students build the same app. The one who explains it clearly to users, listens to feedback and works well in a team usually gets a better outcome than one who only codes.

Lesson at a glance



Discuss with a partner: where do you see 'Skills for the AI Age' in real life? Write your best example.

_____

_____

LESSON 4.3 | PRACTISE: SKILLS FOR THE AI AGE

Chapter 4: AI & Career

Try it

Task
Rate yourself from 1 to 5 in each skill: maths, Python, problem solving, communication, teamwork. Circle your top 2 strengths and 2 skills to grow.

Check your understanding

1. Name three technical skills for AI.

2. Name three human skills.

3. Why can't AI replace human skills like empathy?

4. Why are values important in AI careers?

5. Which skill will you improve first? How?

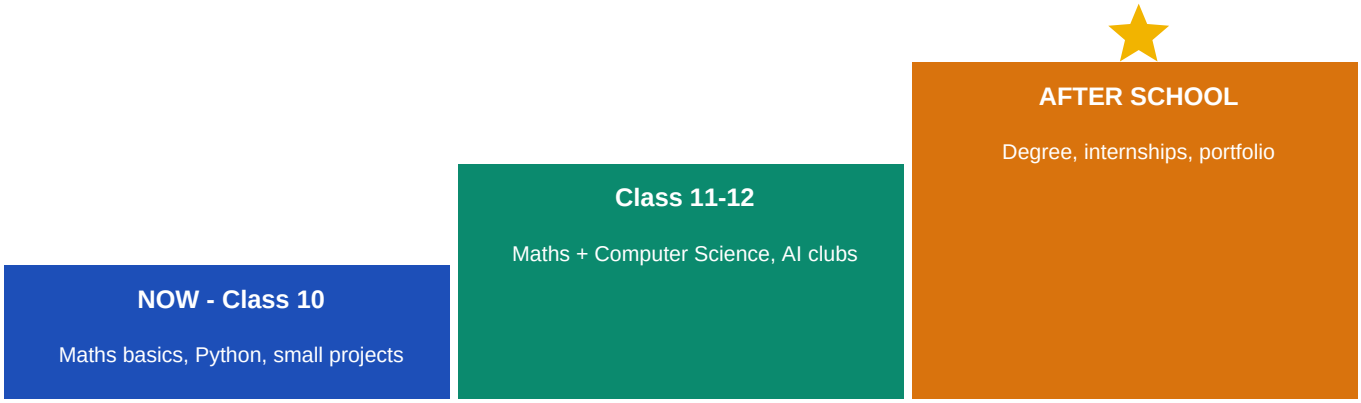
Remember
- AI careers need technical and human skills.
- Both can be practised every day.
- Values and learning habits matter.

LESSON 4.4 | YOUR PATH AFTER CLASS 10

Chapter 4: AI & Career - Learn

After Class 10 you will choose subjects and a stream. Choices made now keep many doors open. Check the latest details with your teacher or counsellor, because rules and courses change over time.

Class 11 and 12	If possible, choose Mathematics and Computer Science or Informatics Practices. Strong maths and logic help in AI.
After school	Degrees in Computer Science, AI and Data Science, Statistics, Electronics or Engineering are common routes. Other fields can add AI skills later.
Other routes	Diplomas, certificate courses and skill programmes also lead to careers in data, coding and AI support work.
Keep options open	Whatever you choose, keep learning programming, data and communication. Plan B routes are also good.



Real-life example
Meena likes biology and computers. She plans Class 11 with Biology, Maths and Computer Science. Later she can study biotechnology or data science and work on AI for health.

Lesson at a glance



Discuss with a partner: where do you see 'Your Path After Class 10' in real life? Write your best example.

_____

_____

LESSON 4.4 | PRACTISE: YOUR PATH AFTER CLASS 10

Chapter 4: AI & Career

Try it

Task
Write a rough plan: which subjects you may choose in Class 11-12 and why. Add one backup plan.

Check your understanding

1. Which two subjects help most for AI?

2. Name two degree areas linked to AI.

3. Why should you keep a backup plan?

4. Where can you check the latest course information?

5. How can you keep learning if your stream is not computer science?

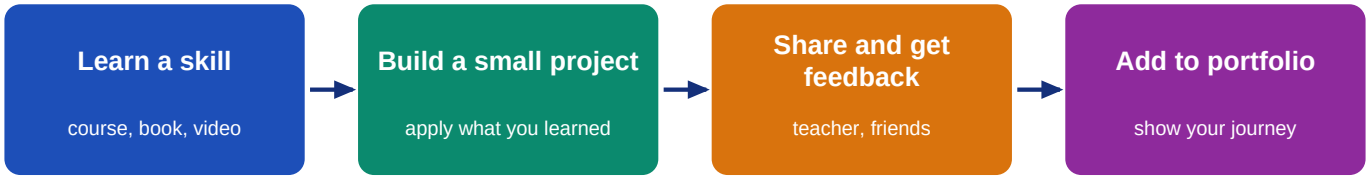
Remember
- Maths and Computer Science help AI careers.
- Many routes lead to AI work.
- Check updated information and keep options open.

LESSON 4.5 | BUILDING SKILLS BEYOND SCHOOL

Chapter 4: AI & Career - Learn

Skills grow with practice. You do not need to wait for college. Learning projects, clubs, free online courses and competitions can help you start today.

Small projects	Build a quiz bot, a marks analyser or a simple image classifier. Start small and finish projects.
Courses and clubs	Join school coding or AI clubs. Use free online courses with your teacher's or parents' guidance.
Portfolio	A portfolio is a collection of your projects with short explanations, photos and what you learned.
Safe learning online	Use trusted sites, protect personal data and never share passwords. Ask an adult before registering for a course.



Real-life example
Karthik makes one small project each month and notes it in a portfolio. After a year he has 12 projects to show in an internship interview.

Lesson at a glance



Discuss with a partner: where do you see 'Building Skills Beyond School' in real life? Write your best example.

_____

_____

LESSON 4.5 | PRACTISE: BUILDING SKILLS BEYOND SCHOOL

Chapter 4: AI & Career

Try it

Task
Plan your next 3 months: one skill to learn, one small project to build and one place to show it (class display, club or school website).

Check your understanding

1. Why are small projects valuable?

2. What is a portfolio?

3. Where can students find learning help safely?

4. Why should you protect your passwords and data online?

5. What will your first project be?

Remember
- Start with small, finished projects.
- Keep a portfolio of your work.
- Learn safely online.

LESSON 4.6 | LEARNING FOR LIFE: FUTURE OF WORK

Chapter 4: AI & Career - Learn

Jobs are changing as AI grows. Some tasks will be automated, some jobs will change and new jobs will appear. The best preparation is to keep learning and to stay human: curious, kind and creative.

Tasks change	Many jobs will change because some tasks are done by AI. People focus more on judgement, creativity and care.
New roles	New jobs appear around data, AI safety, robot maintenance and AI-assisted services.
Lifelong learning	Plan to learn new tools every few years. Learning how to learn is a key skill.
Ethics and trust	Employers value people who use AI honestly, respect privacy and can explain their decisions.

Skills many employers look for (example ranking, not exact data)



Real-life example
A bank clerk now uses AI to read forms quickly. The clerk spends more time helping customers with difficult cases, which needs patience and understanding.

Lesson at a glance



Discuss with a partner: where do you see 'Learning for Life: Future of Work' in real life? Write your best example.

_____

_____

LESSON 4.6 | PRACTISE: LEARNING FOR LIFE: FUTURE OF WORK

Chapter 4: AI & Career

Try it

Task
Interview an adult: How has technology changed their work in the last 10 years? What skill helped them most? Write the answers.

Check your understanding

1. How might AI change jobs?

2. Name one new kind of job created by AI.

3. What is lifelong learning?

4. Why do employers value honesty in AI use?

5. What human skill will you keep strengthening?

Remember
- Jobs change; learning continues.
- Human skills become more valuable.
- Use AI honestly and responsibly.

ACTIVITY 4.1 | SKILL AUDIT

Chapter 4: AI & Career - Do

Time: 20 min **Group:** Individual **Materials:** Worksheet table, pencil

Goal

Understand your current skills and choose two to improve.

Steps

1. Rate yourself from 1 (beginner) to 5 (strong) for each skill listed.

2. Circle your top two strengths.

3. Underline two skills to improve.

4. For each skill to improve, write one action for this month.

5. Share one action with a partner and ask for a suggestion.

Record your results

Skill	My rating (1 to 5)	Action to improve

Think and write

Were you surprised by any rating? Why?

_____

_____

_____

_____

ACTIVITY 4.2 | CAREER INTERVIEW

Chapter 4: AI & Career - Do

Time: Home task + 20 min in class **Group:** Individual **Materials:** 5 prepared questions, notebook

Goal

Learn how technology and AI affect a real job by interviewing a professional.

Steps

1. Choose an adult who uses computers or data at work (parent, teacher, engineer, doctor, bank staff).

2. Prepare 5 questions: What do you do? Which skills matter most? How is AI changing your work? What advice for students? What was your first step?

3. Interview politely and take notes.

4. Write a 5-line summary.

5. Present for 1 minute in class.

Record your results

Question	Answer	What I learned

Think and write

Which advice will you follow first?

_____

_____

_____

_____

ACTIVITY 4.3 | MY CAREER BLUEPRINT

Chapter 4: AI & Career - Do

Time: 30 min **Group:** Individual **Materials:** A4 sheet, colours

Goal

Create a visual roadmap for your next steps.

Steps

1. Draw a 3-step road: Now (Class 10), Class 11-12, After school.

2. On each step write subjects, skills, one project and one course or club.

3. Add a backup plan related to your interests.

4. Decorate your blueprint.

5. Display it and walk around to read others' plans.

Record your results

Stage	Skills and subjects	Project or course

Think and write

What is the first step you will take this month?

_____

_____

_____

_____

ACTIVITY 4.4 | CAREER CARD GAME

Chapter 4: AI & Career - Do

Time: 25 min	**Group:** Groups of 4	**Materials:** Career cards (role + one-line description + 3 skills), blank cards

Goal

Learn what skills each AI career needs.

Steps

1. Deal career cards and skill cards separately.

2. One student reads a career; others hold up the skill cards they think it needs.

3. Check with the card and score a point for each correct skill.

4. Make 2 new career cards of your own (for example AI in cricket, AI in music).

5. Play again with the new cards.

Record your results

Career	Skills named	Score

Think and write

Which career surprised you most?

_____

_____

_____

_____

ACTIVITY 4.5 | PORTFOLIO PAGE DESIGN

Chapter 4: AI & Career - Do

Time: 30 min **Group:** Individual **Materials:** A4 sheet or digital tool

Goal

Learn how to present a project clearly.

Steps

1. Choose one project you have done (Chapter 1, 2 or 3 project).

2. Write its title, the problem it solves and your role.

3. Add a photo, drawing or screenshot.

4. Write 2 lines on what you learned and 1 line on what you would improve.

5. Add your name and date and swap pages for peer feedback.

Record your results

Part of page	What I wrote	Peer suggestion

Think and write

What makes a portfolio page convincing to someone who sees it for the first time?

_____

_____

_____

_____

MINI PROJECT | PART 1: PLAN

Chapter 4: AI & Career

My AI Career Roadmap and Portfolio

Prepare a personal roadmap for the next 5 years and the first page of your portfolio. Include a career or two you are curious about, skills to build and a small project you will complete.

Pick an idea (tick one)

[ ] Roadmap for a Data Analyst career
[ ] Roadmap for an AI in Healthcare career
[ ] Roadmap for a Robotics and AI career
[ ] Roadmap for AI with your own favourite field

Project plan

Plan	My answer
Project name	
Problem I am solving	
Who will use it	
Tools / AI I will use	
Data or materials needed	
Team members and roles	
Time I need	

Success criteria

- Shows at least 2 careers and what they do.
- Lists 5 skills with a plan to build each.
- Includes one finished sample project in the portfolio page.
- Mentions where to get safe, trusted learning help.

MINI PROJECT | PART 2: BUILD, TEST, PRESENT

Chapter 4: AI & Career

Build tip

Use the Skill Audit and Career Interview results. Be realistic and specific, for example 'solve 2 Python problems every week'. Do not share personal details such as phone number or address on a public portfolio.

Test log

Test	What I tried	Result	What I will improve
1			
2			
3			
4			

Presentation checklist (tick when done)

[ ] I can explain the problem in one sentence.
[ ] I can show how my project works.
[ ] I can say where AI or data is used.
[ ] I can name one limit or risk.
[ ] I checked facts and privacy.
[ ] I thanked my team and mentor.

Peer feedback: 2 stars and a wish

Star 1	Star 2	Wish (one improvement)

Project score (teacher)

Idea and purpose	Working prototype	Responsible AI use	Presentation	Total
5	5	5	5	/ 20

WORKSHEET A | CHAPTER 4: AI & CAREER

Name: _____ Class: _____ Date: _____ Marks: ____ / 18

Section A - Choose the correct answer (8 x 1 = 8)

1. Which skill is MOST useful for an AI career?
   (a) Python programming and problem solving
   (b) Only memorising
   (c) Avoiding maths
   (d) Typing speed alone

2. A professional who studies data to find useful insights is a:
   (a) data analyst
   (b) painter
   (c) pilot
   (d) chef

3. Which statement is true?
   (a) AI careers need only coders
   (b) AI careers also need ethics, communication and domain knowledge
   (c) AI careers do not need learning
   (d) AI careers are only for adults

4. The best way to build AI skills beyond school is:
   (a) waiting till college
   (b) regular practice, projects and online learning
   (c) skipping practice
   (d) only reading titles

5. A collection of your projects is called a:
   (a) portfolio
   (b) password
   (c) playlist
   (d) passport

6. Domain knowledge means:
   (a) knowing a website address
   (b) deep knowledge of a field like health or farming
   (c) knowing computer parts
   (d) typing quickly

7. Which of these is a human skill?
   (a) Teamwork
   (b) Compiling code
   (c) Data storage
   (d) Sorting

8. For planning your studies after Class 10 you should:
   (a) ignore updates
   (b) check current information with teachers or counsellors
   (c) follow only rumours
   (d) choose without thinking

Section B - Fill in the blanks (6 x 1 = 6)

1. A collection of your projects that shows your skills is a _____.

2. Two school subjects that support AI learning most are Maths and _____.

3. Learning new things throughout life is called _____ learning.

4. A person who studies AI ethics and safety is an AI _____.

5. Short work experience to learn skills is an _____.

6. A step-by-step plan to reach a goal is a _____.

Section C - True or False (4 x 1 = 4)

1. Only programmers can work in AI. **(True / False)**

2. Human skills like communication are valuable in AI careers. **(True / False)**

3. You should share passwords with online course sites. **(True / False)**

4. Domain knowledge helps spot wrong AI answers. **(True / False)**

WORKSHEET B | CHAPTER 4: AI & CAREER

Name: _____ Class: _____ Date: _____ Marks: ____ / 24

Section D - Match the columns (5 x 1 = 5)

Column A	Answer	Column B
1. Data Analyst	_____	(a) Collection of your projects
2. ML Engineer	_____	(b) People skills like teamwork
3. Portfolio	_____	(c) Finds insights in data
4. Domain knowledge	_____	(d) Deep understanding of a field
5. Soft skills	_____	(e) Builds and runs machine learning systems

Section E - Short answers (4 x 3 = 12)

1. Describe any two AI-related careers and what they do.

2. Which skills do you need to improve, and how will you do it?

3. Why is domain knowledge important when using AI?

4. How can a student build skills beyond school safely?

Section F - My 3-step roadmap (7)

Fill the roadmap table below with your own plan for Class 10, Class 11-12 and after school.

REVISION AND PUZZLE

Chapter 4: AI & Career

This chapter in 10 points

1. AI has many careers: data analyst, data scientist, ML engineer, researcher, robotics engineer, ethicist.

2. AI is used as a tool in almost every field.

3. Domain knowledge plus AI skills is powerful.

4. Technical skills: maths, statistics, Python, data.

5. Human skills: communication, teamwork, creativity, ethics.

6. Choose Maths and Computer Science if you can, and keep options open.

7. Check current course information with teachers or counsellors.

8. Build small projects and keep a portfolio.

9. Learn safely online and protect your data.

10. Jobs change, so keep learning for life.

Word search: find these 8 words (across, down or diagonal)

R	R	A	V	R	O	A	D	M	A	P	O
L	W	L	C	H	M	G	S	D	U	E	D
W	V	J	R	S	K	I	L	L	S	E	U
V	O	R	O	B	O	T	I	C	S	T	C
E	G	F	R	C	O	R	K	R	P	H	Y
M	C	A	R	E	E	R	M	M	Y	I	C
K	H	R	A	N	A	L	Y	S	T	C	K
T	T	P	Y	T	H	O	N	R	L	S	R
C	K	D	V	M	J	M	O	E	C	M	H
B	E	J	P	V	G	A	W	E	U	T	P
A	O	A	H	C	D	N	B	B	I	P	K
K	M	P	O	R	T	F	O	L	I	O	L

CAREER | PORTFOLIO | ANALYST | ROBOTICS | SKILLS | PYTHON | ETHICS | ROADMAP

GLOSSARY AND SELF-CHECK

Chapter 4: AI & Career

Glossary

Data Analyst	Finds insights in data.
Data Scientist	Builds models and studies data to answer questions.
ML Engineer	Builds and runs machine learning systems.
Domain knowledge	Deep understanding of a field like health or farming.
Statistics	Maths of collecting and understanding data.
Portfolio	A collection of your projects.
Internship	Short work experience to learn skills.
Lifelong learning	Learning new skills throughout life.
Roadmap	A step-by-step plan to reach a goal.
Soft skills	People skills such as communication and teamwork.

I can ... (tick the box)

☐	Name and describe AI careers.	Yes / Need help
☐	Explain domain knowledge.	Yes / Need help
☐	List technical and human skills.	Yes / Need help
☐	Make a rough plan for Class 11-12.	Yes / Need help
☐	Create a portfolio page.	Yes / Need help
☐	Explain lifelong learning.	Yes / Need help
☐	Plan small projects to build skills.	Yes / Need help
☐	Learn safely online.	Yes / Need help

My reflection

The most interesting thing I learned in this chapter was ...

_____

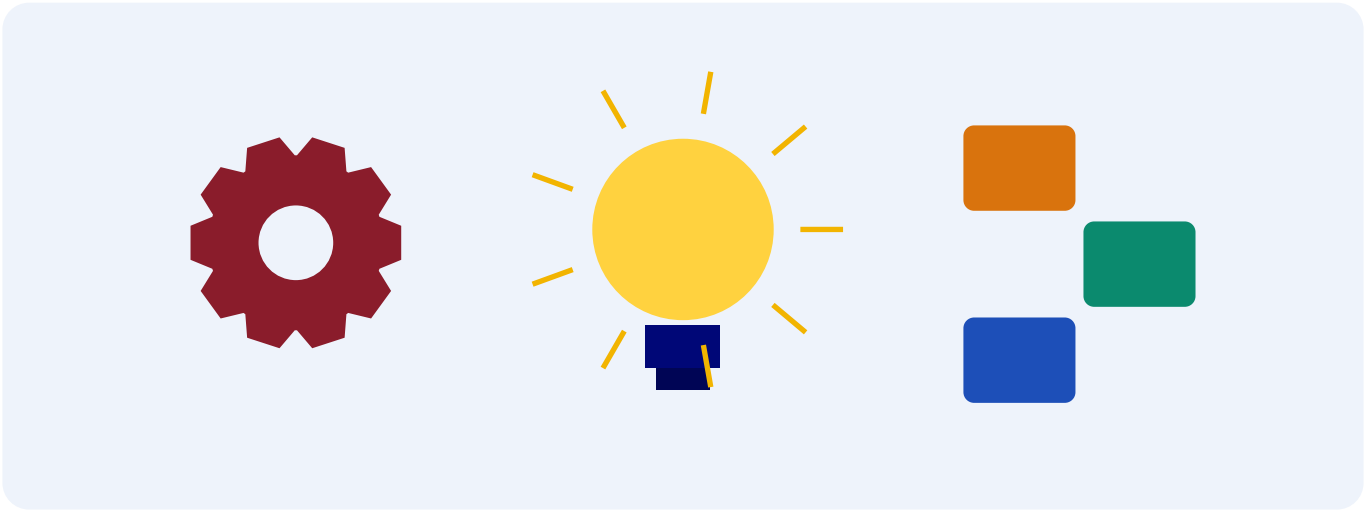
_____

_____

Teacher's remarks: _____ Signature: _____ Date: _____

CHAPTER 5 | MONTH 5 | AI TOOLS

AI Productivity Booster | Create Your AI Project



In this chapter you will

- Choose the right type of AI tool for a task and use it to study and plan better.
- Plan, build, test and present a small AI-based project.
- Verify and improve AI output instead of trusting it blindly.

Key words

Prompt engineering	Writing clear instructions to get useful AI output.
Prototype	An early working version for testing.
Verify	Check that something is true and correct.
No-code tool	A tool that lets you build without writing code.
User testing	Asking real users to try your project.
Productivity	Getting more useful work done in less time.
Summary	A short version that keeps the main points.

Try this
Ask an AI tool to explain a topic to a 10-year-old and then to a Class 10 student. How do the answers differ?

WARM-UP | THINK BEFORE YOU LEARN

Chapter 5: AI Tools

Before you start this chapter, find out what you already know. There are no wrong answers here. Guess, think and write.

My KWL chart

K - What I already KNOW	W - What I WANT to know	L - What I LEARNED (fill at the end)

Guess and write

1. Which apps or tools do you already use to study or create things?

_____

_____

2. What takes you the longest time in your homework or revision?

_____

_____

3. How can you tell if an online answer is correct?

_____

_____

4. Have you ever built or made something from an idea? What was it?

_____

_____

5. Write one project you would love to build this year.

_____

_____

Chapter roadmap

Lesson 5.1
Types of AI Tools

Lesson 5.2
AI as a Study Buddy

Lesson 5.3
Writing, Design and Creativity with AI

Lesson 5.4
Planning Your AI Project

Lesson 5.5
Building and Testing a Prototype

Lesson 5.6
Presenting, Verifying and Improving

LESSON 5.1 | TYPES OF AI TOOLS

Chapter 5: AI Tools - Learn

AI tools are apps and websites that use AI to help you do a task. Different tools are made for different jobs. Choosing the right tool saves time and gives better results.

Chat assistants	Explain topics, summarise text, brainstorm ideas, help with writing and answer follow-up questions.
Image and design tools	Create pictures, posters and icons from a written description. You can edit the result.
Language tools	Translate, check grammar, convert speech to text and text to speech.
Coding and no-code builders	Coding assistants help write programs. No-code tools such as Teachable Machine let you train simple models without writing code.

Chat assistant
explain, summarise

Image tool
poster, picture

Language tool
translate, grammar

Coding helper
write, explain code

No-code builder
train a model

Planner
timetable, tasks

Real-life example
For a school project on water: a chat assistant helps list ideas, an image tool makes a poster draft, a language tool translates the summary into your mother tongue and you check facts in your textbook.

Lesson at a glance



Discuss with a partner: where do you see 'Types of AI Tools' in real life? Write your best example.

_____

_____

LESSON 5.1 | PRACTISE: TYPES OF AI TOOLS

Chapter 5: AI Tools

Try it

Task
List 4 tasks you do for school. Next to each, write the type of AI tool that could help and one thing you must still do yourself.

_____

_____

_____

_____

_____

Check your understanding

1. What is an AI tool?

_____

_____

2. Name three types of AI tools.

_____

_____

3. What is a no-code tool? Give an example.

_____

_____

4. Why should you choose the right tool for the task?

_____

_____

5. What should you still do yourself while using AI tools?

_____

_____

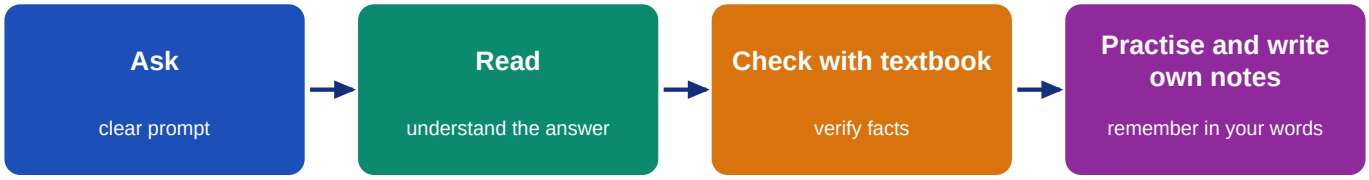
Remember
- Different AI tools do different jobs.
- Choose the tool that fits the task.
- You stay in charge of thinking and checking.

LESSON 5.2 | AI AS A STUDY BUDDY

Chapter 5: AI Tools - Learn

Used well, AI is like a patient study partner. It can explain, quiz and plan with you. Used badly, it becomes a shortcut that stops you from learning. The difference is how you use it.

Understand	Ask for a simple explanation or an example. Say 'explain like I am in Class 8' if the first answer is too hard.
Summarise	Paste your own notes (no private data) and ask for a 6-point summary. Then compare it with the textbook.
Practise	Ask for 5 practice questions with answers. Try to solve them first without looking at the answers.
Plan	Ask for a 7-day revision timetable using your subjects and free hours, then adjust it yourself.



Real-life example
Before a science test, Priya asks AI to explain 'reflection of light' with a mirror example. She then writes her own notes, solves 5 questions and checks them with her textbook.

Lesson at a glance



Discuss with a partner: where do you see 'AI as a Study Buddy' in real life? Write your best example.

_____

_____

LESSON 5.2 | PRACTISE: AI AS A STUDY BUDDY

Chapter 5: AI Tools

Try it

Task
Choose a topic you find hard. Write the prompt you would give an AI study buddy to (a) explain it, (b) quiz you and (c) plan 3 days of revision.

Check your understanding

1. Give two good ways to use AI for studying.

2. What should you do after AI summarises a chapter?

3. Why try the practice questions before seeing the answers?

4. What is wrong with copying AI answers into homework?

5. Write one prompt for a revision timetable.

Remember
- Use AI to understand, practise and plan.
- Verify with your textbook and write in your own words.
- AI is a study buddy, not a homework machine.

LESSON 5.3 | WRITING, DESIGN AND CREATIVITY WITH AI

Chapter 5: AI Tools - Learn

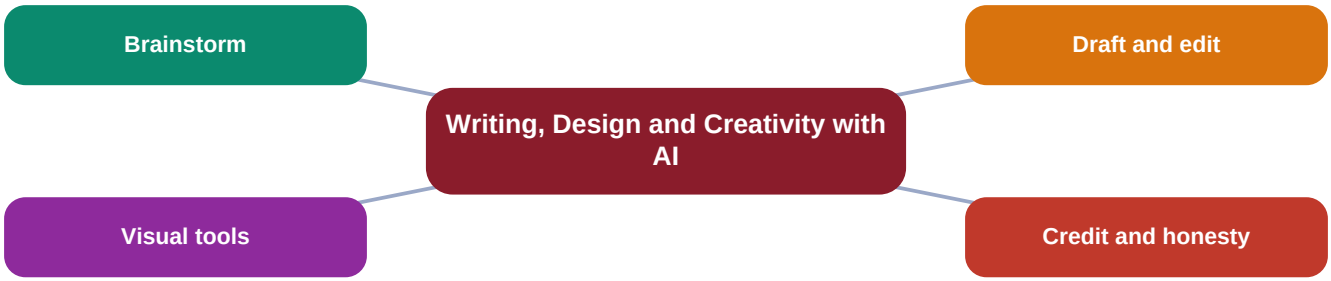
AI can spark ideas and make first drafts of stories, posters, slides and music. But the final work should be yours: your ideas, your edits and your voice.

Brainstorm	Ask for 10 ideas on a topic, then pick two and improve them.
Draft and edit	Let AI draft an outline. You rewrite in your own words, add real examples and correct facts.
Visual tools	Describe the picture you want: subject, style, colours, mood. Refine with follow-up prompts.
Credit and honesty	Say when AI helped. Do not claim AI-made content as fully your own, and respect copyright.

AI does well	You must do
• Many quick ideas • First drafts • Variations of a design • Language polish	• Choose the best idea • Check facts • Add your voice • Give credit and be honest

Real-life example
A student asks AI for 10 slogans for 'Save Water', picks the best two, changes a few words and adds a personal story. The final poster is clearly the student's own creation.

Lesson at a glance



Discuss with a partner: where do you see 'Writing, Design and Creativity with AI' in real life? Write your best example.

_____

_____

LESSON 5.3 | PRACTISE: WRITING, DESIGN AND CREATIVITY WITH AI

Chapter 5: AI Tools

Try it

Task
Ask an AI tool (or imagine its answer) for 5 slogans on 'Reduce Plastic'. Choose one and rewrite it better. Explain what you changed.

Check your understanding

1. How can AI help in brainstorming?

2. Why should you edit AI drafts?

3. What details help an image tool make a better picture?

4. Why must you give credit when AI helped?

5. Who owns the final idea: AI or you? Explain.

Remember
- AI helps with ideas and drafts.
- You edit, check and add your own voice.
- Give credit and be honest.

LESSON 5.4 | PLANNING YOUR AI PROJECT

Chapter 5: AI Tools - Learn

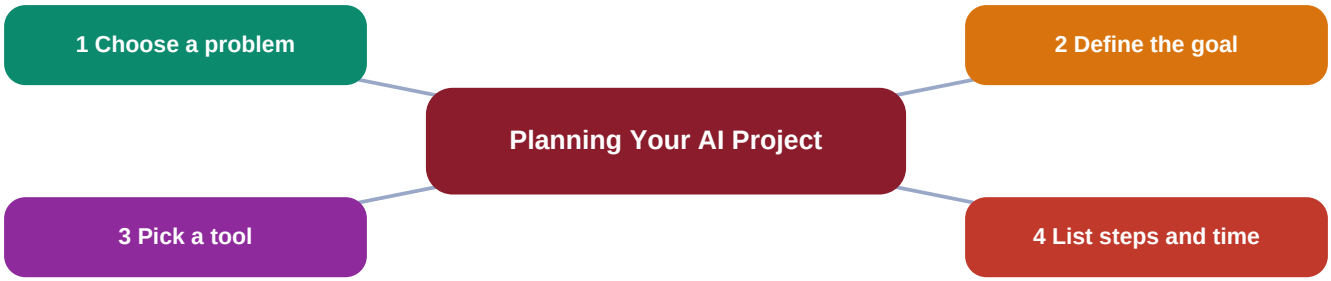
A project turns an idea into something real that can be tested. Good projects solve a clear problem for real users and are small enough to finish.

1 Choose a problem	Pick something real and local. Ask: who has this problem, how often and why?
2 Define the goal	Write one sentence: 'My project helps ___ to ___ by using ___'.
3 Pick a tool	Choose the simplest tool that can do the job: chat assistant, no-code model builder, spreadsheet or Python.
4 List steps and time	Break the project into small tasks with deadlines. Assign roles in your team.



Real-life example
Goal: 'My project helps classmates revise by using an AI-made quiz from our own notes.' Tool: chat assistant. Steps: collect notes, write prompts, test quiz, ask 5 classmates for feedback.

Lesson at a glance



Discuss with a partner: where do you see 'Planning Your AI Project' in real life? Write your best example.

_____

_____

LESSON 5.4 | PRACTISE: PLANNING YOUR AI PROJECT

Chapter 5: AI Tools

Try it

Task
Write your project goal in one sentence using the pattern above. Then list 5 small steps with a date for each.

Check your understanding

1. Why should a project solve a real problem?

2. Write the goal pattern for a project.

3. How do you decide which tool to use?

4. Why break a project into small tasks?

5. What roles can team members take?

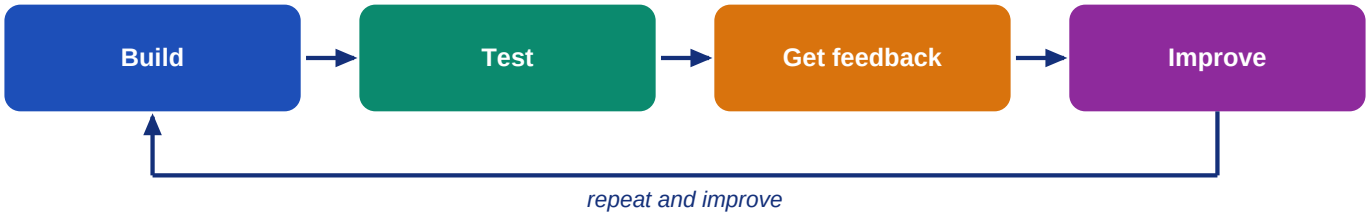
Remember
- Start from a real problem.
- Write a clear goal sentence.
- Choose the simplest tool and break work into steps.

LESSON 5.5 | BUILDING AND TESTING A PROTOTYPE

Chapter 5: AI Tools - Learn

A prototype is an early working version of your project. It does not need to be perfect. Its job is to be tested so that you can learn what to improve.

Build the simplest version	Make just enough to test the main idea, such as one quiz, one classifier or one chatbot answer.
Collect good data	Use varied, correct and permitted data. Do not use private information or photos of people without permission.
Test with real users	Ask classmates to try it. Watch what they do and note where they get confused.
Record and improve	Keep a test log: what you tried, what happened and what you will change. Then improve and test again.



Real-life example
A team builds an image model for 'plastic vs paper'. It works on clean photos but fails on dirty items. They add dirty-item photos and the accuracy improves.

Lesson at a glance



Discuss with a partner: where do you see 'Building and Testing a Prototype' in real life? Write your best example.

_____

_____

LESSON 5.5 | PRACTISE: BUILDING AND TESTING A PROTOTYPE

Chapter 5: AI Tools

Try it

Task
Plan 3 tests for your prototype: who will test, what will they do, and what will you measure (for example how many answers were correct).

Check your understanding

1. What is a prototype?

2. Why should a prototype be tested with real users?

3. What kind of data should you avoid?

4. What goes into a test log?

5. Why do we repeat build, test and improve?

Remember
- Prototype means early version for testing.
- Use permitted and varied data.
- Test, record, improve and repeat.

LESSON 5.6 | PRESENTING, VERIFYING AND IMPROVING

Chapter 5: AI Tools - Learn

A great project must be explained clearly and checked carefully. Verify facts, state limits honestly and invite feedback. Good presenters also credit tools and sources.

Tell the story	Problem, idea, how it works, results and what you learned. Use simple words and one demo.
Verify	Check numbers and claims in trusted sources. Say which parts AI helped with and which you checked yourself.
State limits	Say honestly where the project may fail and what you will do next.
Use feedback	Listen, note the suggestions and improve. Feedback is a gift, not an attack.

Presentation part	What to include
Problem	Who has it and why it matters
Solution	What you built and how it works
Evidence	Test results and sources
Limits and next steps	Where it fails and what you will improve

Real-life example
A group says: 'Our quiz maker was correct for 18 of 20 questions. Two answers were wrong, so we checked every question with the textbook. Next, we will add a hint feature.'

Lesson at a glance



Discuss with a partner: where do you see 'Presenting, Verifying and Improving' in real life? Write your best example.

_____

_____

LESSON 5.6 | PRACTISE: PRESENTING, VERIFYING AND IMPROVING

Chapter 5: AI Tools

Try it

Task
Write a 60-second presentation script for your project idea using the four parts in the table.

Check your understanding

1. What should a good presentation include?

2. Why must you state the limits of your project?

3. How do you verify AI-made content?

4. What should you do with feedback?

5. Why credit the tools and sources you used?

Remember
- Explain problem, solution, evidence and limits.
- Verify and credit.
- Use feedback to improve.

ACTIVITY 5.1 | 30-MINUTE PRODUCTIVITY CHALLENGE

Chapter 5: AI Tools - Do

Time: 30 min **Group:** Individual **Materials:** One textbook chapter, AI tool or teacher-made AI output

Goal

Use AI to study faster while still checking and thinking for yourself.

Steps

1. Ask AI to summarise the chapter in 8 points.

2. Ask for 5 practice questions with answers.

3. Check each answer against the textbook and mark mistakes.

4. Make a 3-day revision plan.

5. Compare plans with a classmate and improve yours.

Record your results

AI output	Correct? (Yes / No)	What I fixed

Think and write

Which part saved the most time? Which part needed the most checking?

_____

_____

_____

_____

ACTIVITY 5.2 | TOOL MATCH AND CHOOSE

Chapter 5: AI Tools - Do

Time: 20 min **Group:** Pairs **Materials:** Task cards and tool cards

Goal

Choose the right AI tool for different tasks.

Steps

1. Read 8 task cards, such as 'translate a notice' or 'make a poster'.

2. Match each with a tool type card.

3. Explain your choice to your partner.

4. Write one task where no AI tool is the best choice.

5. Share one surprising match with the class.

Record your results

Task	Tool type	Why this tool

Think and write

When is it better not to use AI at all?

_____

_____

_____

_____

ACTIVITY 5.3 | PROJECT IDEA PITCH

Chapter 5: AI Tools - Do

Time: 25 min **Group:** Groups of 3 **Materials:** Project planning sheet, timer

Goal

Brainstorm and select a realistic project idea.

Steps

1. Brainstorm 3 ideas.

2. Score each idea 1 to 5 for useful, possible and safe.

3. Choose the best idea and fill the plan sheet.

4. Pitch it in 90 seconds.

5. Collect two questions and answer them.

Record your results

Idea	Useful / Possible / Safe	Total score

Think and write

What question from the class helped you most?

_____

_____

_____

_____

ACTIVITY 5.4 | PROTOTYPE AND PEER TEST

Chapter 5: AI Tools - Do

Time: 40 min **Group:** Groups of 3 **Materials:** Computer or AI tool, test log

Goal

Build a simple prototype and test it with another team.

Steps

1. Build a simple prototype (a quiz, an image model, a chatbot prompt or a small program).

2. Another team tests it and gives 2 positives and 2 improvements.

3. Record everything in the test log.

4. Improve the prototype and test again.

5. Note what changed and why.

Record your results

Test	Result	Improvement made

Think and write

What did the other team notice that you had missed?

_____

_____

_____

_____

ACTIVITY 5.5 | VERIFY IT! FACT-CHECK SPRINT

Chapter 5: AI Tools - Do

Time: 25 min **Group:** Pairs **Materials:** 6 AI-written statements, textbook or trusted website

Goal

Practise checking AI answers with trusted sources.

Steps

1. Get 6 AI-written statements from the teacher.

2. Predict which ones are correct.

3. Check each in a textbook or trusted website.

4. Mark correct, wrong or unclear.

5. Rewrite wrong statements correctly.

Record your results

Statement no.	Correct / Wrong	Source used

Think and write

How many predictions were right? What will you do before trusting an AI answer?

_____

_____

_____

_____

MINI PROJECT | PART 1: PLAN

Chapter 5: AI Tools

Create Your AI Project

Plan and build a small AI-powered project that solves a real problem. Show your plan, prototype, test log, results and presentation.

Pick an idea (tick one)

[ ] Class notes quiz maker
[ ] School FAQ chatbot using a no-code tool
[ ] Plant leaf classifier (healthy vs damaged) using a no-code tool
[ ] Python marks analyser that prints average, highest and lowest

Project plan

Plan	My answer
Project name	
Problem I am solving	
Who will use it	
Tools / AI I will use	
Data or materials needed	
Team members and roles	
Time I need	

Success criteria

- Clear problem and goal sentence.
- Working prototype tested with at least 5 users.
- Test log with improvements made.
- Facts verified and AI help credited.

MINI PROJECT | PART 2: BUILD, TEST, PRESENT

Chapter 5: AI Tools

Build tip

Keep the first version very small. Use permitted data only. Test with 5 classmates and record exactly what they did. Note one limit of your project.

Test log

Test	What I tried	Result	What I will improve
1			
2			
3			
4			

Presentation checklist (tick when done)

[ ] I can explain the problem in one sentence.
[ ] I can show how my project works.
[ ] I can say where AI or data is used.
[ ] I can name one limit or risk.
[ ] I checked facts and privacy.
[ ] I thanked my team and mentor.

Peer feedback: 2 stars and a wish

Star 1	Star 2	Wish (one improvement)

Project score (teacher)

Idea and purpose	Working prototype	Responsible AI use	Presentation	Total
5	5	5	5	/ 20

WORKSHEET A | CHAPTER 5: AI TOOLS

Name: _____ Class: _____ Date: _____ Marks: ____ / 18

Section A - Choose the correct answer (8 x 1 = 8)

1. The best use of an AI tool for studying is to:
   (a) copy answers without reading
   (b) summarise a chapter and check it with the textbook
   (c) skip lessons
   (d) ask it to attend exams

2. The first step in creating an AI project is to:
   (a) choose and define a problem
   (b) publish the project
   (c) choose a colour theme
   (d) delete old files

3. A prototype is:
   (a) the final unchangeable product
   (b) an early working version used for testing
   (c) a type of virus
   (d) a keyboard

4. Before using AI output in homework you should:
   (a) submit directly
   (b) verify with reliable sources
   (c) add more mistakes
   (d) hide it

5. A no-code tool lets you:
   (a) build without writing code
   (b) only write code
   (c) only watch videos
   (d) only print pages

6. A test log records:
   (a) what you tried, result and improvement
   (b) only your name
   (c) only the date
   (d) nothing

7. When AI helped you, you should:
   (a) hide it
   (b) say so and give credit
   (c) delete the history
   (d) blame the AI

8. Which data should you avoid in a project?
   (a) varied permitted examples
   (b) private photos of people without permission
   (c) your own drawings
   (d) public domain pictures

Section B - Fill in the blanks (6 x 1 = 6)

1. Writing clear instructions to get good AI output is called prompt _____.

2. A first trial version of a project is a _____.

3. AI output must always be _____ before use.

4. Comments from users that help you improve are called _____.

5. Telling people whose work or tools you used is giving _____.

6. Getting more useful work done in less time is called _____.

Section C - True or False (4 x 1 = 4)

1. AI tools can replace your own thinking completely. **(True / False)**

2. You should test a prototype with real users. **(True / False)**

3. It is fine to claim AI-made work as fully your own. **(True / False)**

4. A project goal should be small enough to finish. **(True / False)**

WORKSHEET B | CHAPTER 5: AI TOOLS

Name: _____ Class: _____ Date: _____ Marks: ____ / 24

Section D - Match the columns (5 x 1 = 5)

Column A	Answer	Column B
1. Prototype	_____	(a) Build without writing code
2. Verify	_____	(b) Improving again and again
3. No-code tool	_____	(c) Early working version for testing
4. User testing	_____	(d) Asking real users to try the project
5. Iteration	_____	(e) Check that something is true

Section E - Short answers (4 x 3 = 12)

1. List 3 ways AI tools can improve your study productivity.

2. Describe your project idea in 3 lines.

3. Why should you test a prototype with real users?

4. How do you make sure AI-generated information is correct?

Section F - Project planning sheet (7)

Fill in the plan for your AI project in the table below.

REVISION AND PUZZLE

Chapter 5: AI Tools

This chapter in 10 points

1. AI tools are made for different jobs: chat, images, language, coding and no-code building.

2. Use AI as a study buddy to understand, practise and plan.

3. Verify AI answers with textbooks and trusted sources.

4. AI can help brainstorm and draft; you edit and add your own voice.

5. Give credit and be honest about AI help.

6. Projects start from real problems with clear goals.

7. Break projects into small steps and roles.

8. A prototype is an early version used for testing.

9. Test with real users, keep a log and improve.

10. Present problem, solution, evidence and limits.

Word search: find these 8 words (across, down or diagonal)

M	G	V	E	R	I	F	Y	I	H	S	R
M	O	G	E	O	D	O	U	S	G	T	Y
A	U	L	D	E	S	I	G	N	P	N	T
I	H	G	L	L	B	S	F	B	L	O	K
J	D	C	M	T	P	U	B	B	A	M	N
P	R	O	T	O	T	Y	P	E	N	A	L
C	R	E	D	I	T	I	A	E	N	E	Y
O	P	D	J	K	O	M	D	A	E	T	S
C	E	L	R	J	S	U	B	G	R	K	O
Y	F	E	E	D	B	A	C	K	V	L	M
N	C	T	E	S	T	I	N	G	B	D	R
T	L	C	B	Y	G	P	R	O	M	P	T

PROTOTYPE | VERIFY | PROMPT | FEEDBACK | PLANNER | CREDIT | DESIGN | TESTING

GLOSSARY AND SELF-CHECK

Chapter 5: AI Tools

Glossary

Prompt engineering	Writing clear instructions to get useful AI output.
Prototype	An early working version for testing.
Verify	Check that something is true and correct.
No-code tool	A tool that lets you build without writing code.
User testing	Asking real users to try your project.
Productivity	Getting more useful work done in less time.
Summary	A short version that keeps the main points.
Iteration	Improving again and again.
Feedback	Comments that help you improve.
Credit	Telling people whose work or tools you used.

I can ... (tick the box)

☐	Choose a suitable AI tool for a task.	Yes / Need help
☐	Use AI to study and make a timetable.	Yes / Need help
☐	Edit AI drafts into my own work.	Yes / Need help
☐	Write a clear project goal.	Yes / Need help
☐	Plan steps and roles.	Yes / Need help
☐	Build and test a prototype.	Yes / Need help
☐	Keep a test log and improve.	Yes / Need help
☐	Present my project honestly.	Yes / Need help

My reflection

The most interesting thing I learned in this chapter was ...

_____

_____

_____

Teacher's remarks: _____ Signature: _____ Date: _____

CHAPTER 6 | MONTH 6 | RESPONSIBLE AI

Original or AI-Made? | Use AI, Don't Misuse AI



In this chapter you will

- Spot signs of AI-generated content, deepfakes and misinformation.
- Understand academic honesty, privacy, bias and copyright when using AI.
- Create personal and class rules for safe, honest and fair use of AI.

Key words

Deepfake	AI-made or altered media that looks real.
Misinformation	False or misleading information.
Plagiarism	Passing off someone else's work as your own.
Disclosure	Telling others that AI helped you.
Privacy	Keeping personal details safe.
Bias	Unfair results caused by one-sided data.
Copyright	Legal ownership of creative work.

Think about it
If a video looks real, does that mean it is real? What would you check first?

WARM-UP | THINK BEFORE YOU LEARN

Chapter 6: Responsible AI

Before you start this chapter, find out what you already know. There are no wrong answers here. Guess, think and write.

My KWL chart

K - What I already KNOW	W - What I WANT to know	L - What I LEARNED (fill at the end)

Guess and write

1. Have you ever seen a photo or video online that you later found out was fake? What happened?

_____

_____

2. Is it fair to submit homework written by someone else? Why or why not?

_____

_____

3. What personal details should never be shared online?

_____

_____

4. Who should decide the rules for using AI at school?

_____

_____

5. Write one rule you would make for using AI responsibly.

_____

_____

Chapter roadmap

Lesson 6.1
Original or AI-Made?

Lesson 6.2
Deepfakes and Misinformation

Lesson 6.3
Academic Honesty and AI

Lesson 6.4
Privacy and Safety

Lesson 6.5
Bias, Fairness and Copyright

Lesson 6.6
Use AI, Don't Misuse AI: Our Rules

LESSON 6.1 | ORIGINAL OR AI-MADE?

Chapter 6: Responsible AI - Learn

AI can now write essays, draw pictures and copy voices. Sometimes it is hard to know what is real. Learning to look carefully, ask questions and check sources protects you and others.

Signs in text	Very generic wording, repeated phrases, confident but wrong facts, and no personal touch can be signs of AI writing.
Signs in images	Odd hands or fingers, garbled letters, strange shadows, blurred backgrounds or perfect-looking skin may point to AI images.
Signs in audio and video	Voice that sounds flat, lips not matching speech, odd blinking or lighting changes may be clues.
Clues are not proof	No single clue proves anything. AI-detector tools can also be wrong. Always check the source and context.

Possible AI signs	Better checks
• Generic, repeated wording • Garbled text in pictures • Odd hands or shadows • Lips not matching voice	• Who posted it? Is it trusted? • Does the same news appear elsewhere? • Search the image or quote • Ask a teacher or adult

Real-life example
A class receives a photo of a 'new giant bridge'. The text on the signboard is garbled and no news site reports it. The class concludes it is probably AI-made.

Lesson at a glance



Discuss with a partner: where do you see 'Original or AI-Made?' in real life? Write your best example.

_____

_____

LESSON 6.1 | PRACTISE: ORIGINAL OR AI-MADE?

Chapter 6: Responsible AI

Try it

Task
Look at two pieces of media provided by your teacher (or find two safe examples). Write the clues you notice and one question you would ask before believing each.

_____

_____

_____

_____

_____

Check your understanding

1. Name two clues that an image may be AI-made.

_____

_____

2. Why is no single clue proof?

_____

_____

3. Why are AI detectors not fully reliable?

_____

_____

4. What is a better check than only looking at a picture?

_____

_____

5. Who can you ask if you are not sure?

_____

_____

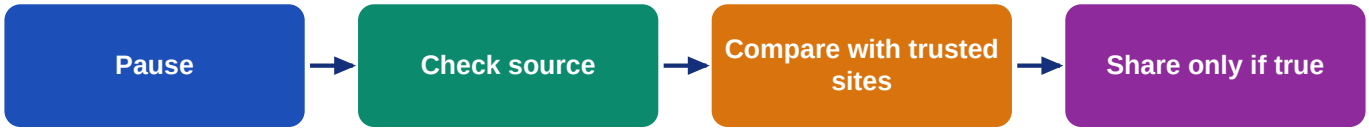
Remember
- Look for clues but do not rely on one clue.
- Check the source and other reports.
- AI detectors can be wrong.

LESSON 6.2 | DEEPFAKES AND MISINFORMATION

Chapter 6: Responsible AI - Learn

A deepfake is a video, image or audio made or changed by AI to look real. Misinformation is false or misleading information. Together they can hurt people and spread panic.

What is a deepfake	AI may place a person's face or voice into content they never created. It can be used for fun, but also for lies and bullying.
Harm	Fake videos can damage reputation, cause fear, cheat people out of money and spread rumours quickly.
Pause and check	Stop before you share. Check who posted it, look for the same news on trusted sites and look for signs of editing.
Act responsibly	Do not forward unverified content. Report harmful fakes on the platform and tell a trusted adult or teacher.



Real-life example
A fake voice message pretending to be a school principal announces a holiday. Students who checked with the school office found it false and avoided confusion.

Lesson at a glance



Discuss with a partner: where do you see 'Deepfakes and Misinformation' in real life? Write your best example.

_____

_____

LESSON 6.2 | PRACTISE: DEEPFAKES AND MISINFORMATION

Chapter 6: Responsible AI

Try it

Task
Write a 4-step 'Before I share' checklist you will follow whenever you see shocking news or a video.

Check your understanding

1. What is a deepfake?

2. What is misinformation?

3. Why is pausing before sharing important?

4. What should you do if a fake targets a classmate?

5. Why check more than one trusted source?

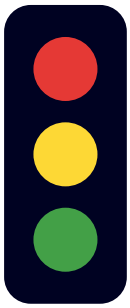
Remember
- Deepfakes are AI-made or changed media.
- Pause, check and compare before sharing.
- Report harmful fakes and tell an adult.

LESSON 6.3 | ACADEMIC HONESTY AND AI

Chapter 6: Responsible AI - Learn

Using AI to learn is smart. Using AI to cheat is dishonest. Academic honesty means that the work you submit shows your own learning, and that you tell the truth about any help you received.

Allowed use	Ask AI to explain, quiz you, give feedback on your draft or check grammar, if your teacher allows.
Not allowed	Submitting AI-written answers as your own, copying friends' work or using AI during tests where it is banned.
Plagiarism	Passing off someone else's words or ideas as yours, whether from a person, a website or AI.
Disclose	Tell your teacher how AI helped: 'I used AI to check my grammar and to get quiz questions.' Honesty builds trust.



RED - Misuse
Cheating, fake media, sharing private data

YELLOW - Not sure
Ask your teacher or parent first

GREEN - Responsible use
Learning, practice, checking your work, with honesty

Real-life example
Ravi writes his own essay, asks AI to point out unclear sentences and rewrites them himself. He adds a note: 'AI used for feedback only.' This is honest and helps him learn.

Lesson at a glance



Discuss with a partner: where do you see 'Academic Honesty and AI' in real life? Write your best example.

_____

_____

LESSON 6.3 | PRACTISE: ACADEMIC HONESTY AND AI

Chapter 6: Responsible AI

Try it

Task
Write 3 examples of using AI that are allowed in your school and 3 that are not allowed. Mark any you are unsure about with a question mark.

Check your understanding

1. What is academic honesty?

2. Give two allowed uses of AI in studies.

3. What is plagiarism?

4. Why should you disclose AI help?

5. What can you do when you are not sure whether AI use is allowed?

Remember
- Use AI to learn, not to cheat.
- Follow school rules and be honest.
- Disclose how AI helped you.

LESSON 6.4 | PRIVACY AND SAFETY

Chapter 6: Responsible AI - Learn

Every message, photo and form you share online can leave a trail called your digital footprint. AI tools may store what you type. Protecting your privacy protects you and your family.

Personal data	Full name with school, address, phone number, ID numbers, passwords, photos of others and family details are private.
Think before typing	Do not paste private information into AI tools or apps. Use fake sample data in practice work.
Settings and permissions	Check app permissions for camera, location and microphone. Allow only what is needed.
Safe behaviour	Use strong passwords, do not share them, log out on shared devices and tell an adult if something feels unsafe.

Safe to share	Do not share
Your favourite subject	Phone number and address
A general question	Passwords and OTPs
Your own artwork (with care)	ID numbers and bank details
Fake sample names for practice	Photos of others without permission

Real-life example
A student pastes a full class list with phone numbers into an AI tool to make a timetable. This shares others' private details. The student should have used fake sample data.

Lesson at a glance



Discuss with a partner: where do you see 'Privacy and Safety' in real life? Write your best example.

_____

_____

LESSON 6.4 | PRACTISE: PRIVACY AND SAFETY

Chapter 6: Responsible AI

Try it

Task
Check one app on your phone or a school computer (with permission). List what permissions it asks for and which ones you think are not needed.

Check your understanding

1. What is a digital footprint?

2. Name four types of private data.

3. Why use fake sample data for practice?

4. What are app permissions?

5. What should you do if someone online asks for your OTP?

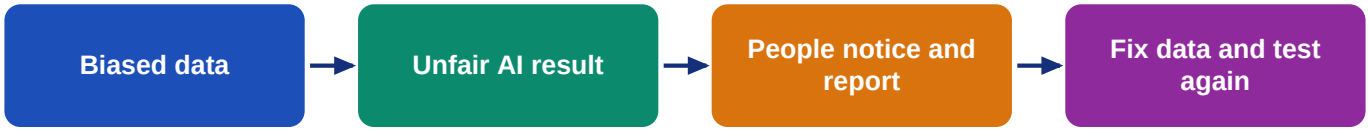
Remember
- Protect personal data online.
- Do not paste private information into AI tools.
- Tell an adult when something feels unsafe.

LESSON 6.5 | BIAS, FAIRNESS AND COPYRIGHT

Chapter 6: Responsible AI - Learn

AI learns from data made by people, so it can copy human bias. It also learns from creative work made by people, which raises questions about ownership and credit. Responsible users think about both.

What is bias	If data over-represents or stereotypes some groups, AI results can be unfair. Example: a hiring tool that prefers one kind of applicant.
Reducing bias	Use varied data, test with different groups, let humans review important decisions and report unfair results.
Copyright	Writers, artists and musicians own their work. Do not copy it and claim it as yours. Check usage rules.
Credit and permission	Say which tools or sources you used. Ask permission before using someone's photo, voice or artwork.



Real-life example
A voice assistant understands some accents better than others because it was trained mostly on those accents. Adding more varied voices improves fairness.

Lesson at a glance



Discuss with a partner: where do you see 'Bias, Fairness and Copyright' in real life? Write your best example.

_____

_____

LESSON 6.5 | PRACTISE: BIAS, FAIRNESS AND COPYRIGHT

Chapter 6: Responsible AI

Try it

Task
Think of an AI tool you use. List two groups of people who may be treated differently by it. Write one test you could do to find out.

Check your understanding

1. What is AI bias?

2. How can bias be reduced?

3. What is copyright?

4. Why ask permission before using someone's photo?

5. Why should humans review important AI decisions?

Remember
- AI can copy human bias.
- Use varied data and human review.
- Respect copyright and give credit.

LESSON 6.6 | USE AI, DON'T MISUSE AI: OUR RULES

Chapter 6: Responsible AI - Learn

Now you know the risks and the good uses. The final step is to make your own rules. Responsible AI users are curious, honest, careful and kind.

Use	Learning, brainstorming, practice, translation, accessibility, checking your own work and building projects.
Misuse	Cheating, creating fake photos or voices of others, bullying, spreading lies and sharing private data.
Four checks	Is it honest? Is it safe? Is it fair? Is it kind? If any answer is 'no', stop and ask.
Speak up	If you see misuse, tell a teacher or trusted adult. Helping to keep the community safe is part of responsible AI.

Honest
Say when AI helped

Safe
Protect private data

Fair
Check for bias

Kind
Never harm or tease others

Verified
Check before you share

Learning
Think for yourself

Real-life example
A class writes a pledge: 'We use AI to learn, we verify facts, we protect privacy and we respect each other.' They put it on the wall and review it every month.

Lesson at a glance



Discuss with a partner: where do you see 'Use AI, Don't Misuse AI: Our Rules' in real life? Write your best example.

_____

_____

LESSON 6.6 | PRACTISE: USE AI, DON'T MISUSE AI: OUR RULES

Chapter 6: Responsible AI

Try it

Task
Write 5 rules for your own AI use. Start each with 'I will' and make them specific.

_____

_____

_____

_____

_____

Check your understanding

1. Give two examples of responsible use of AI.

_____

_____

2. Give two examples of misuse.

_____

_____

3. What are the four checks?

_____

_____

4. What should you do if you see misuse?

_____

_____

5. Which rule from your list is most important to you and why?

_____

_____

Remember
- Use AI to learn and create.
- Ask: honest, safe, fair, kind.
- Speak up when you see misuse.

ACTIVITY 6.1 | REAL OR AI? SPOT THE DIFFERENCE

Chapter 6: Responsible AI - Do

Time: 20 min **Group:** Teams of 4 **Materials:** 8 samples (images or text, real and AI-made) prepared by the teacher

Goal

Practise looking for clues and checking sources.

Steps

1. Look at each sample for 1 minute.

2. List clues that make you think it is real or AI-made.

3. Vote Real or AI as a team.

4. The teacher reveals the answers.

5. Discuss which clues helped and which misled you.

Record your results

Sample no.	Our vote and clues	Actual answer

Think and write

Which clue was the least reliable? What would you check next time?

_____

_____

_____

_____

ACTIVITY 6.2 | GREEN LIGHT, YELLOW LIGHT, RED LIGHT

Chapter 6: Responsible AI - Do

Time: 20 min **Group:** Whole class **Materials:** 10 scenario cards, green, yellow and red cards

Goal

Decide quickly and fairly whether an AI use is responsible, uncertain or a misuse.

Steps

1. The teacher reads a scenario, for example 'Using AI to explain a maths step'.

2. Hold up green (responsible), yellow (ask first) or red (misuse).

3. Explain your choice in one sentence.

4. Discuss any scenario where the class disagrees.

5. Write 3 scenarios of your own.

Record your results

Scenario	Colour I chose	Reason

Think and write

Which scenario was hardest to judge? What rule would help?

_____

_____

_____

_____

ACTIVITY 6.3 | FACT-CHECK DETECTIVE

Chapter 6: Responsible AI - Do

Time: 30 min **Group:** Pairs **Materials:** 3 claims or viral messages (teacher provides), trusted websites or books

Goal

Practise checking viral claims using trusted sources.

Steps

1. Read each claim carefully.

2. List what you would check: source, date, other reports.

3. Search for the claim in at least two trusted sources.

4. Decide: true, false or unclear.

5. Write a short message you could send if the claim is false.

Record your results

Claim	Sources checked	True / False / Unclear

Think and write

What clue showed that a claim could be false? How will you stop false news from spreading?

_____

_____

_____

_____

ACTIVITY 6.4 | PRIVACY CHECK-UP

Chapter 6: Responsible AI - Do

Time: 25 min	**Group:** Individual then pairs	**Materials:** Checklist, fake sample profile, school computer or tablet

Goal

Learn to spot private data and safe settings.

Steps

1. Read a fake sample profile and mark the private details.

2. List which details should never be shared.

3. Check the privacy settings of a demo app with your teacher.

4. Write 3 safe habits.

5. Share your habits with a partner.

Record your results

Detail in profile	Private? (Yes / No)	What to do instead

Think and write

Which habit will you start today? Whom will you tell about it?

_____

_____

_____

_____

ACTIVITY 6.5 | CLASS AI PLEDGE

Chapter 6: Responsible AI - Do

Time: 30 min | **Group:** Groups of 5, then class | **Materials:** Chart paper, markers

Goal

Create a shared set of rules for responsible AI use.

Steps

1. Each group writes 5 rules for using AI responsibly.

2. Share rules and combine them into one class pledge.

3. Vote on the final 8 rules.

4. Design the pledge poster.

5. Everyone signs and the poster is displayed.

Record your results

Rule	Why it matters	Who agrees (initials)

Think and write

Which rule will be hardest to follow? How can the class help each other?

_____

_____

_____

_____

MINI PROJECT | PART 1: PLAN

Chapter 6: Responsible AI

Responsible AI Awareness Campaign

Create a campaign for your school or community that teaches safe and honest AI use. It may be a poster set, a short skit, a video script or an assembly talk. Include real examples, tips and a pledge.

Pick an idea (tick one)

[ ] Poster series: Spot the Deepfake
[ ] Skit: The Fake Message that Went Viral
[ ] Assembly talk: Use AI, Don't Misuse AI
[ ] Pocket guide: 10 Rules for Responsible AI

Project plan

Plan	My answer
Project name	
Problem I am solving	
Who will use it	
Tools / AI I will use	
Data or materials needed	
Team members and roles	
Time I need	

Success criteria

- Accurate facts and examples.
- Clear tips on privacy, honesty and verification.
- No real person's photo or voice used without permission.
- Includes a call to action or pledge.

MINI PROJECT | PART 2: BUILD, TEST, PRESENT

Chapter 6: Responsible AI

Build tip

Use made-up or teacher-approved samples, never real students' photos. Test your message with 5 listeners and ask what they remember. Credit any tools used.

Test log

Test	What I tried	Result	What I will improve
1			
2			
3			
4			

Presentation checklist (tick when done)

[ ] I can explain the problem in one sentence.
[ ] I can show how my project works.
[ ] I can say where AI or data is used.
[ ] I can name one limit or risk.
[ ] I checked facts and privacy.
[ ] I thanked my team and mentor.

Peer feedback: 2 stars and a wish

Star 1	Star 2	Wish (one improvement)

Project score (teacher)

Idea and purpose	Working prototype	Responsible AI use	Presentation	Total
5	5	5	5	/ 20

WORKSHEET A | CHAPTER 6: RESPONSIBLE AI

Name: _____ Class: _____ Date: _____ Marks: ____ / 18

Section A - Choose the correct answer (8 x 1 = 8)

1. Which is a possible sign that an image is AI-made?
   (a) Clear natural hands
   (b) Garbled text and odd hands or shadows
   (c) A printed date
   (d) A visible camera brand

2. Submitting an AI-written assignment as your own work is:
   (a) good practice
   (b) academic dishonesty
   (c) a compulsory rule
   (d) a new subject

3. What should you NOT share with an AI chatbot?
   (a) A general science question
   (b) Your phone number, address and ID numbers
   (c) A maths problem
   (d) A poem topic

4. A deepfake is:
   (a) a very deep lake
   (b) AI-made or altered media that looks real
   (c) a type of battery
   (d) a slow computer

5. Before sharing shocking news you should:
   (a) share immediately
   (b) pause and check trusted sources
   (c) delete your phone
   (d) add more details

6. Bias in AI often comes from:
   (a) one-sided or incomplete data
   (b) fast internet
   (c) bright screens
   (d) long passwords

7. Copyright protects:
   (a) only computers
   (b) the creative work of its owner
   (c) only old books
   (d) nothing

8. If you see a harmful fake about a classmate you should:
   (a) forward it
   (b) report it and tell a trusted adult
   (c) laugh
   (d) ignore everything

Section B - Fill in the blanks (6 x 1 = 6)

1. Checking a claim with several trusted sources is called _____ checking.

2. When AI treats some groups unfairly because of its training data, it shows _____.

3. Telling others that AI helped you with your work is _____ AI use.

4. The trail of data you leave online is your digital _____.

5. False or misleading information is called _____.

6. Passing off someone else's work as your own is _____.

Section C - True or False (4 x 1 = 4)

1. AI detector tools are always correct. **(True / False)**

2. You should pause and check before sharing a shocking video. **(True / False)**

3. It is fine to share a classmate's phone number with an AI tool. **(True / False)**

4. Using AI to get feedback on your own draft can be responsible if allowed. **(True / False)**

WORKSHEET B | CHAPTER 6: RESPONSIBLE AI

Name: _____ Class: _____ Date: _____ Marks: ____ / 24

Section D - Match the columns (5 x 1 = 5)

Column A	Answer	Column B
1. Deepfake	_____	(a) Keeping personal details safe
2. Plagiarism	_____	(b) Telling others that AI helped you
3. Privacy	_____	(c) AI-made or altered media that looks real
4. Copyright	_____	(d) Legal ownership of creative work
5. Disclosure	_____	(e) Passing off others' work as yours

Section E - Short answers (4 x 3 = 12)

1. Give 2 examples of responsible use of AI and 2 examples of misuse.

2. You receive a shocking video of a public figure on social media. What steps will you take?

3. Why is protecting privacy important when using AI tools?

4. How can AI bias be reduced?

Section F - Write your Class AI Pledge (7)

Write 5 rules you will follow while using AI. Start each rule with 'I will'.

REVISION AND PUZZLE

Chapter 6: Responsible AI

This chapter in 10 points

1. AI can create text, images and voices that look real.

2. Clues help, but no single clue is proof; check source and context.

3. Deepfakes and misinformation can harm people.

4. Before sharing: pause, check, compare, then decide.

5. Use AI to learn; submitting AI work as your own is cheating.

6. Disclose when AI helped you.

7. Protect private data and use fake sample data for practice.

8. AI can copy human bias, so use varied data and human review.

9. Respect copyright and give credit.

10. Ask: Is it honest, safe, fair and kind?

Word search: find these 8 words (across, down or diagonal)

E	B	I	A	S	V	F	W	N	U	C	E
C	M	M	U	O	R	I	S	A	W	D	T
W	O	S	I	V	K	D	J	N	F	O	H
A	I	P	D	E	E	P	F	A	K	E	I
P	R	F	Y	R	D	V	J	V	R	U	C
G	R	P	E	R	L	F	S	R	V	A	S
U	K	I	L	P	I	A	D	L	E	J	H
B	H	T	V	E	C	G	C	P	R	C	S
E	E	W	P	A	D	S	H	F	I	I	R
U	N	G	S	Y	C	G	G	T	F	Y	J
M	W	V	L	O	R	Y	E	O	Y	D	H
H	C	K	A	T	S	H	O	N	E	S	T

DEEPFAKE | PRIVACY | BIAS | HONEST | COPYRIGHT | VERIFY | ETHICS | PLEDGE

GLOSSARY AND SELF-CHECK

Chapter 6: Responsible AI

Glossary

Deepfake	AI-made or altered media that looks real.
Misinformation	False or misleading information.
Plagiarism	Passing off someone else's work as your own.
Disclosure	Telling others that AI helped you.
Privacy	Keeping personal details safe.
Bias	Unfair results caused by one-sided data.
Copyright	Legal ownership of creative work.
Fact-checking	Checking a claim with trusted sources.
Digital footprint	The trail of data you leave online.
Ethics	Rules of right and wrong behaviour.

I can ... (tick the box)

☐	Spot clues of AI-made content.	Yes / Need help
☐	Explain deepfakes and misinformation.	Yes / Need help
☐	Follow Pause, Check, Compare, Share.	Yes / Need help
☐	Explain academic honesty and disclosure.	Yes / Need help
☐	Protect my personal data online.	Yes / Need help
☐	Explain AI bias and how to reduce it.	Yes / Need help
☐	Respect copyright and give credit.	Yes / Need help
☐	Make and follow my own AI pledge.	Yes / Need help

My reflection

The most interesting thing I learned in this chapter was ...

_____

_____

_____

Teacher's remarks: _____ Signature: _____ Date: _____

MY AI LEARNING JOURNAL

Fill this at the end of every month

Month	What I learned	My project / challenge	Rating (1-5)
1. AI Basics			
2. Communicate with AI			
3. AI Applications			
4. AI & Career			
5. AI Tools			
6. Responsible AI			

My goal for the next 6 months:

_____

_____

_____

_____

ANSWER KEY (For Teachers)

Lesson check questions: answers are in the lesson text; accept correct answers in the student's own words. Sample answers are given for open questions.

Chapter 1: AI Basics

A (MCQ): 1-(b), 2-(b), 3-(b), 4-(b), 5-(b), 6-(a), 7-(b), 8-(b)
B (Fill): 1. patterns; 2. hallucination; 3. supervised; 4. token; 5. tested; 6. bias
C (True/False): 1-False, 2-True, 3-False, 4-True
D (Match): 1-(c), 2-(e), 3-(a), 4-(d), 5-(b) [Data = Facts, text, images or numbers AI learns from; Model = The trained 'brain' that makes predictions; Prompt = Instruction given to an AI tool; Hallucination = Confident but false AI answer; Reinforcement learning = Learning by rewards and penalties]
E1: Sample: The filter is shown thousands of emails labelled spam or not spam. It finds patterns (such as 'free prize', unknown senders) and uses them to predict if a new email is spam.
E2: Sample: It can hallucinate false facts. It can show bias from training data. (Also accept: no real understanding, outdated knowledge.)
E3: Sample: Classification puts items in groups, such as spam or not spam. Prediction estimates a value or event, such as tomorrow's temperature.
E4: Sample: AI learns only from data. Poor, wrong or one-sided data gives poor or unfair results (garbage in, garbage out).
F: Sample: ear shape, snout length, body size. May fail for puppies, kittens or unusual breeds. More varied photos reduce such failures.

Chapter 2: Communicate with AI

A (MCQ): 1-(b), 2-(b), 3-(b), 4-(b), 5-(c), 6-(b), 7-(b), 8-(b)
B (Fill): 1. role; 2. edge; 3. debugging; 4. variable; 5. input; 6. refining
C (True/False): 1-False, 2-True, 3-True, 4-False
D (Match): 1-(c), 2-(e), 3-(a), 4-(d), 5-(b) [Role = Who the AI should act as; Context = Background information for the task; Format = How the answer should look; Constraint = A limit such as word count; Bug = A mistake in a program]
E1: Sample: 'Act as a Python teacher. Write a program that asks for two numbers and prints their sum. Use simple names and add comments.'
E2: Sample: It may contain bugs or unsafe code, you will not learn, and you cannot fix or explain it.
E3: Sample: An unusual input that may break the program, such as 0 for a division program or an empty list.
E4: Sample: Code - get a first version and read it. Test - run with normal and edge inputs. Improve - fix errors and test again.
F: Bug: range(1, 5) stops at 4. Fix: range(1, 6). Prompt example: 'This code should print 15 but prints 10. Explain the bug and show the fix.'

Chapter 3: AI Applications

A (MCQ): 1-(b), 2-(b), 3-(b), 4-(c), 5-(a), 6-(b), 7-(b), 8-(b)
B (Fill): 1. traffic; 2. quality; 3. data; 4. risk; 5. stakeholder; 6. automation
C (True/False): 1-False, 2-True, 3-False, 4-True
D (Match): 1-(c), 2-(e), 3-(a), 4-(d), 5-(b) [Fraud detection = Spotting unusual or false transactions; Chatbot = Program that talks with people; Sensor = Device that measures moisture, heat or motion; Stakeholder = Person or group affected by a decision; Prediction = Estimate of what is likely to happen]
E1: Sample: Food waste in the canteen. AI could predict how many meals are needed from past data.
E2: Sample: A wrong prediction could harm a patient or private data could leak. A doctor must verify and data must be protected.
E3: Sample: 1 Problem, 2 Data, 3 AI method, 4 Benefit, 5 Risk. Each step is described briefly.
E4: Sample: AI can be wrong or biased. A human can check, correct and take responsibility.

F: Sample: Problem = pest damage; Data = leaf photos, weather, soil; AI method = image recognition; Benefit = early treatment, saved crops; Risk = wrong detection, needs training and expert check.

Chapter 4: AI & Career

A (MCQ): 1-(a), 2-(a), 3-(b), 4-(b), 5-(a), 6-(b), 7-(a), 8-(b)
B (Fill): 1. portfolio; 2. Computer Science; 3. lifelong; 4. ethicist; 5. internship; 6. roadmap
C (True/False): 1-False, 2-True, 3-False, 4-True
D (Match): 1-(c), 2-(e), 3-(a), 4-(d), 5-(b) [Data Analyst = Finds insights in data; ML Engineer = Builds and runs machine learning systems; Portfolio = Collection of your projects; Domain knowledge = Deep understanding of a field; Soft skills = People skills like teamwork]
E1: Sample: Data Analyst studies data to find insights. ML Engineer builds and runs AI models.
E2: Answers vary. Look for specific actions, for example 'solve 1 Python problem daily'.
E3: Sample: It helps ask the right questions and spot wrong AI answers.
E4: Sample: Do small projects, join school clubs, use trusted free courses with adult guidance, protect passwords and personal data.
F: Answers vary. Check for clear subject choice, a skill, a project and a timeline at each stage.

Chapter 5: AI Tools

A (MCQ): 1-(b), 2-(a), 3-(b), 4-(b), 5-(a), 6-(a), 7-(b), 8-(b)
B (Fill): 1. engineering; 2. prototype; 3. verified / checked; 4. feedback; 5. credit; 6. productivity
C (True/False): 1-False, 2-True, 3-False, 4-True
D (Match): 1-(c), 2-(e), 3-(a), 4-(d), 5-(b) [Prototype = Early working version for testing; Verify = Check that something is true; No-code tool = Build without writing code; User testing = Asking real users to try the project; Iteration = Improving again and again]
E1: Sample: summarise notes, create practice questions, make a revision timetable, explain tough topics step by step.
E2: Answers vary. Look for a clear problem, a tool and an expected benefit.
E3: Sample: Users find confusing parts and mistakes that the builders miss.
E4: Sample: Check with textbook or trusted sources, compare with other sources and ask a teacher.
F: Answers vary. Check that each box is specific and that the test plan includes real users.

Chapter 6: Responsible AI

A (MCQ): 1-(b), 2-(b), 3-(b), 4-(b), 5-(b), 6-(a), 7-(b), 8-(b)
B (Fill): 1. fact; 2. bias; 3. disclosing; 4. footprint; 5. misinformation; 6. plagiarism
C (True/False): 1-False, 2-True, 3-False, 4-True
D (Match): 1-(c), 2-(e), 3-(a), 4-(d), 5-(b) [Deepfake = AI-made or altered media that looks real; Plagiarism = Passing off others' work as yours; Privacy = Keeping personal details safe; Copyright = Legal ownership of creative work; Disclosure = Telling others that AI helped you]
E1: Sample: Responsible - practising quiz questions; checking grammar. Misuse - copying an AI essay; creating fake photos of a classmate.
E2: Sample: Do not forward. Check trusted news sites. Look for signs of editing. Report it on the platform. Tell a teacher or adult if it targets someone.
E3: Sample: Private data can be stored, misused or leaked, which can harm you and others.
E4: Sample: Use varied data, test with different groups, keep human review and report unfair results.
F: Answers vary. Look for honesty, privacy, verification, credit and respect for others.